

“IoT based food cold storage monitoring and controlling system”

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Electrical Engineering**

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CERTIFICATE

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DECLARATION

We, hereby declare that the Project seminar report “**IoT based food cold storage monitoring and controlling system**” Submitted here

in has been carried out by us in the department of ELECTRICAL ENGINEERING of GURU NANAK INSTITUTE OF ENGINEERING & TECHNOLOGY, NAGPUR. This work is original and not submitted earlier as a whole or in part for the award of any degree/ diploma at this or other Institution/ University.

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This work is dedicated to my

*Beloved Parents,
Respected Teachers and
The Almighty*

Who have been sources of inspiration throughout my life

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ABSTRACT

Tricholoma matsutake (*T. matsutake*) is a special type of fungus known as “the king of bacteria”, and has the very high economic value. However, it is also very difficult to transport due to its corruptibility. Therefore, tracing and tracking the quality and safety of *T. matsutake* in the cold chain is very important and necessary. Based on changes in the cold chain environmental parameters determine the safety of *T. matsutake* is a viable option. This paper developed and tested a real-time monitoring traceability system (RM-TM) using emerging Internet of Things (IoT) technologies for monitoring the cold chain logistics environmental parameters of *T. matsutake*. Finally, system testing and evaluation have shown that RM-TM can track and monitor temperature, humidity, oxygen and carbon dioxide fluctuations in the cold chain in real-time. In addition, the collected data can be used to increase the transparency of cold chain logistics and to more effectively control quality, safety, and traceability. In general, the system evaluation results show that it is reliable and meets the requirements of users.

In the Energy Management system, the main constraints are accurate data, energy monitoring and implementation of visual data for consumer. This Project is intended in designing a system at home or industry which monitors the temperature consumption of cold storage, which is designed to calculate the total energy consumption. A server will be created with appropriate channels to monitor the energy consumption from each of the devices respectively. These data will be uploaded to the server at the monitoring end. Considering all these data individual energy load profile for each of the devices is displayed on the web-page.

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For Ex.:

A

AC	Alternating Current
ASAI	Automated System America Inc
ASUI	Average service Unavailability Index
AEMO	Australian Energy Market Operator
AVR	Automatic Voltage Regulation

C

CAPEX	Capital Expenditure
CO ₂	Carbon dioxide
CST	Concentrated Solar Thermal
CAIFI	Customer Average Interruption Frequency Index
CAIDI	Customer Average Interruption Duration Index
CSP	Concentrating Solar Power
CST	Concentrating Solar Thermal

D

DC	Direct Current
DG	Distributed Generation
DISREL	Distribution System Reliability Evaluation
DDG	Distributed Decentralized Generation
DCF	Discounted Cash Flow

DPP Digital Photo Professional

E

EPRI Electric Power Research Institute

EMS Energy Management System

EIA Energy Information Administration

EPA Environmental Protection Agency

EC European Commission

EPS Electric Power System

ESTELA European Solar Thermal Electricity Association

EBIT Earnings Before Interest

EU European Union

EPIA European Photovoltaic Industry Association

ESTELA European Solar Thermal Electricity Association

ETAP Electrical Transient Analyzer Program

G

GOI Government of India

GW Giga Watt

GWEC Global Wind Energy Council

G-S Gauss-Seidel

GIS Geographical Information System

H

HPR Heat to Power Ratio

I

IEEE	Institute of Electrical and Electronics Engineers
IEA	International Energy Agency
ILS	Intelligent Load Shedding

K

KV	Kilo-Volt
KW	Kilo-Watt

L

LOCE	Levelised Cost of Energy
LTC	Load Tap Changing
LRMC	Long Run Marginal Cost

M

MW	Mega Watt
min	Minimum
MATLAB	Matrix Laboratory
MVA	Mega Volt Ampere
MOSPI	Ministry and Status of Development and Industrial Programmes

P

PU	Per Unit
PM	Particulate Matters
PV	Photovoltaic

PV Present Value

R

RGGVY Rajiv Gandhi Grameen Vidyutikaran Yojana

S

SO₂ Sulphur-di-oxide

SAIFI System Average Interruption Frequency Index

SAIDI System Average Interruption Duration Index

U

U.S. United State

Symbols

For Ex.:

$V_i^{(0)}$	Initial Voltage
V_i	Voltage in ith Iteration
V	System Voltage
$\delta_i^{(0)}$	Phase Angle
$P_{i(sch)}$	Scheduled Active Power
$P_i^{(k)}$	Active Power
$\Delta P_i^{(k)}$	Residual Active Power
P_{gt}	Generated Active Voltage
P_{dt}	Active demand Voltage
$Q_{i(sch)}$	Scheduled Reactive Power
$Q_i^{(k)}$	Reactive Power
$\Delta Q_i^{(k)}$	Residual Reactive Power
Q_{gt}	Generated Reactive Voltage
Q_{dt}	Reactive demand Voltage
S_{gt}	Total Generated Power
S	Total power
I	Total Current
ϵ	Specified Accuracy
J_1	1 st coefficient of Jaccobian Matrix
J_2	2 nd coefficient of Jaccobian Matrix
J_3	3 rd coefficient of Jaccobian Matrix
J_4	4 th coefficient of Jaccobian Matrix

CHAPTER 1

Introduction

CHAPTER No. 1

INTRODUCTION

In recent years, the Internet of Things (IoT) has become an important research topic [7–11], and the emerging IoT technologies are gradually ripening. After nearly a decade of development, the definition of the IoT has become more extensive, and the application of IoT involves more fields that include supply chain management, health care, utilities, transportation and other field. In general, the IoT includes two important features: one is to collect data and connect itself to the cloud's perceived.

A warehouse is a commercial building for storage of goods. Warehouses are used by manufacturers, importers, exporters, wholesalers, transport businesses, customs, etc. Strauss Frito-Lay uses a lot of warehouses to store large amount ofVegetables/Fruits. These fruits and vegetables (Primarily Potatoes) require a certain temperature to be maintained for proper storage. When the chiller or the compressor is switched ON, the RH (Relative Humidity) in the air starts diminishing. It also needs to be monitored else the potatoes will rot sooninthe absence of water content in the air. To overcome these issues, a Wireless Sensor Node based design is proposed which will be able to monitor various sections of the warehouse. It then sends the information to a central node in the warehouse which will collect the data and push it on the cloud.

Food is considered as one of the essential thing for our lives. It is important to reduce food waste and increase the production-consumption ratio. The continuous growth in the cold chain management has been reflected in the growth of globalization. Eighteen per cent of India's fruit and vegetable production, valued at INR 133 billion, is wasted annually. Shortage of government owned cold storages and cold storages owned by mostly the upper-class people in towns, making it unavailable for the poor or low class farmers. Moreover, a dearth of continuous electricity, absence of any warning systems, and labor power add to

the troubles of cold storage owners. Evidently there is a lack of comprehensive technology for the entire produce range of vegetables, fruits, flowers, meat and cereals. Naturally food begins to spoil the moment it is harvested. Micro-organisms which are responsible for spoiling the food cannot survive at extreme temperatures. Food can be saved at very high and very low temperatures. It is tedious to attain higher temperatures and therefore we basically choose the later one to store food. Farmers struggle and work hard to produce the crops and if we cannot utilize or consume it fully, all the work of farmers will go into vain. A proper storage mechanism should be incorporated to avoid the food wastage. Demand can irrespective of seasons which in turn avoids fluctuation in price of the product. Currently many warehouses are deployed where the crops can be stored and this system lacks transparency in the entire process. Farmers are getting looted by the third parties who buy the crops at very low price compared with market price. This is a major loop hole in the legacy system and needs to be overcome. Full control over a product from incubation stage to delivery stage should be given to the person who is sole responsible for the product. Col chain management plays a vital role in storing the products at sustainable environment. Cold- chain is a temperature specific supply chain. It enhances the shelf life of the products such agricultural produce, dairy products, sea food, pharmaceutical drugs, chemicals etc. A perfect cold chain includes production, storage, distribution activities which should be desirably done under low temperatures. Storage is considered as one of the important task in the cold chain. Cold storages need to be monitored continuously to avoid unwelcomed situations. A proper alarm system should be deployed in the cold storages which alerts the users at bizarre conditions. Monitoring can be done by using a platform which displays the status of the cold storage. Sensors can be used to sense the environment and this data can be presented to the users. Temperature can be controlled by the users.

We are building the wireless sensor network based temperature monitoring node

which will send data to the cloud server that we can monitor online.

The Internet of Things (IoT) is becoming more widely used technology nowadays. It is often used to refer to the growing network for connected devices, or “things”, that are capable of exchange data over on a low bandwidth network. IoT is being used in various areas, such as automotive industry, logistics, healthcare ,smart grid and smart cities.

1.1 Background:-

The project background of an IoT-based cold storage monitoring and controlling system is to provide an efficient and effective way of monitoring and controlling the temperature and humidity levels in cold storage facilities. The system utilizes IoT devices such as sensors, actuators, and microcontrollers to collect and transmit data to a central server. The data collected is then analyzed to provide insights into the performance of the cold storage facility.

The project aims to address the challenges faced by cold storage facility managers in maintaining optimal temperature and humidity levels. Traditional cold storage monitoring systems are often manual, which can be time-consuming and prone to errors. An IoT-based system provides real-time monitoring and alerts facility managers of any deviations from the optimal levels.

The system also provides remote access to the cold storage facility, allowing facility managers to control the temperature and humidity levels from anywhere. This feature is particularly useful for large facilities or facilities located in remote areas.

The project background also includes the development of a user-friendly interface that allows facility managers to easily monitor and control the cold storage environment. The interface provides real-time data visualization and alerts, making it easy for facility managers to make informed decisions about the facility's performance.

Overall, the project aims to improve the efficiency and effectiveness of cold storage facilities by utilizing IoT technology to provide real-time monitoring and control.

1.2. Impact of Social Expectation:-

Social expectations of an IoT-based cold storage monitoring and controlling system can have a significant impact on the adoption and success of the system. Some of the social expectations that may impact the system include:

1. Environmental Impact: There is a growing expectation for businesses to adopt sustainable practices and reduce their environmental impact. An IoT-based cold storage system can help reduce energy consumption and minimize waste by optimizing the temperature and humidity levels in the facility. This can improve the social perception of the business and enhance its reputation.

2. Food Safety: Consumers are increasingly concerned about food safety and expect businesses to take measures to ensure the safety of their products. An IoT-based cold storage system can help businesses maintain optimal temperature and humidity levels, reducing the risk of spoilage and contamination. This can improve consumer confidence in the business and enhance its reputation.

3. Data Privacy: With the increasing use of IoT devices, there is a growing concern about data privacy and security. Consumers expect businesses to take measures to protect their personal data and ensure that it is not misused. An IoT-based cold storage system should have robust security measures in place to protect the data collected and transmitted by the system.

4. Efficiency and Productivity: Businesses are expected to adopt technologies that improve efficiency and productivity. An IoT-based cold storage system can automate many of the tasks involved in monitoring and controlling the cold storage environment, reducing the workload for employees and improving overall

efficiency.

Overall, social expectations can have a positive impact on the adoption and success of an IoT-based cold storage monitoring and controlling system. By addressing these expectations, businesses can improve their reputation, enhance consumer confidence, and improve their overall efficiency and productivity.

1.3. Problem Statement:-

The problem statement for an IoT-based cold storage monitoring and controlling system can be stated as follows:

Cold storage units are used in various industries to store perishable items such as food products, pharmaceuticals, and chemicals. Maintaining the optimal temperature and humidity levels inside the cold storage unit is crucial for preserving the quality and safety of these items. However, manual monitoring and control of the cold storage unit can be challenging, time-consuming, and prone to errors. Fluctuations in temperature and humidity levels can lead to spoilage and wastage of the stored items, resulting in significant financial losses for the industry.

Therefore, there is a need for an automated monitoring and controlling system that can provide real-time data on the temperature and humidity levels inside the cold storage unit and adjust them automatically to maintain optimal conditions. An IoT-based solution can provide a cost-effective and efficient way to monitor and control the cold storage unit, ensuring that the stored items are always at optimal conditions and reducing the risk of spoilage and wastage.

1.4 Propose of Study:-

The purpose of studying an IoT-based cold storage monitoring and controlling system can be to:

1. Improve the efficiency and safety of cold storage operations: An IoT-based solution can provide real-time monitoring and control of the cold storage unit, ensuring that the temperature and humidity levels are always at optimal conditions. This can help to prevent spoilage and wastage of perishable items, reduce the risk of accidents, and improve the overall efficiency and safety of cold storage operations.
2. Reduce the cost of cold storage operations: Manual monitoring and control of the cold storage unit can be time-consuming and prone to errors, leading to increased costs for the industry. An IoT-based solution can provide a cost-effective and efficient way to monitor and control the cold storage unit, reducing the need for manual labor and minimizing the risk of errors.
3. Improve the quality and safety of stored items: Maintaining the optimal temperature and humidity levels inside the cold storage unit is crucial for preserving the quality and safety of stored items such as food products, pharmaceuticals, and chemicals. An IoT-based solution can provide real-time data on the temperature and humidity levels inside the cold storage unit and adjust them automatically to maintain optimal conditions, ensuring that the stored items are always at optimal conditions and reducing the risk of spoilage and contamination.
4. Develop new technologies and applications: Studying an IoT-based cold

storage monitoring and controlling system can help to develop new technologies and applications in the field of IoT and cold storage operations. This can lead to the development of new sensors, microcontrollers, cloud platforms, and user interfaces that can be used in other industries and applications.

1.4. Technology of Base:-

An IoT-based cold storage monitoring and controlling system involves the use of various technologies to monitor and control the temperature and humidity levels inside the cold storage unit. Here are some of the key technologies involved in an IoT-based cold storage monitoring and controlling system:

1. Sensors: Sensors are used to monitor the temperature and humidity levels inside the cold storage unit. These sensors can be of different types, such as temperature sensors, humidity sensors, and pressure sensors. The sensors send data to the microcontroller, which processes the data and sends it to the cloud platform for storage and analysis.

2. Microcontroller: A microcontroller is used to process the data received from the sensors and control the temperature and humidity levels inside the cold storage unit. The microcontroller can be programmed to adjust the temperature and humidity levels based on the data received from the sensors. It can also send alerts and notifications to the user interface in case of any abnormal conditions.

3. Cloud platform: A cloud platform is used to store and analyze the data received from the sensors and microcontroller. The cloud platform can provide real-time data analysis, reporting, and visualization of the temperature and humidity levels inside the cold storage unit. It can also provide remote access to the system, allowing users to monitor and control the system from anywhere.

4. Connectivity: Connectivity is crucial for an IoT-based cold storage monitoring and controlling system. The system requires a reliable and secure network connection to transmit data between the sensors, microcontroller, cloud platform, and user interface. Various connectivity options can be used, such as Wi-Fi, Bluetooth, cellular, or Ethernet.

5. User interface: A user interface is used to provide a graphical representation of the temperature and humidity levels inside the cold storage unit. The user interface can be accessed through a web-based application or a mobile application. It can provide real-time alerts and notifications, historical data analysis, and remote control of the system.

1.5. Sensors use in IOT-based cold storage monitoring and controlling system:-

Sensors are an essential component of an IoT-based cold storage monitoring and controlling system. They are used to measure the temperature, humidity, and other environmental parameters inside the cold storage unit. Here are some of the sensors commonly used in an IoT-based cold storage monitoring and controlling system:

1. Temperature sensors: Temperature sensors are used to measure the temperature inside the cold storage unit. There are different types of temperature sensors such as thermocouples, resistance temperature detectors (RTDs), and thermistors. These sensors can measure the temperature accurately and send the data to the microcontroller.

2. Humidity sensors: Humidity sensors are used to measure the relative humidity inside the cold storage unit. There are different types of humidity

sensors such as capacitive, resistive, and thermal conductivity sensors. These sensors can measure the humidity accurately and send the data to the microcontroller.

3. Pressure sensors: Pressure sensors are used to measure the pressure inside the cold storage unit. These sensors can detect any changes in pressure, which can indicate a problem with the cooling system or a leak in the cold storage unit.

4. Gas sensors: Gas sensors are used to detect any harmful gases or chemicals inside the cold storage unit. These sensors can detect gases such as ammonia, carbon dioxide, and methane, which can be hazardous to human health.

5. Motion sensors: Motion sensors are used to detect any movement inside the cold storage unit. These sensors can detect any unauthorized access to the cold storage unit and send alerts to the user interface.

6. Door sensors: Door sensors are used to detect the opening and closing of the cold storage unit door. These sensors can detect any unauthorized access to the cold storage unit and send alerts to the user interface.

CHAPTER 2

Literature Review

CHAPTER No. 2

LITERATURE REVIEW

1. **“ Smart Temperature Monitoring System Using ESP32 and DS18B20 ” Santoso Budijono, Felita, Computer Engineering Department, Faculty of Engineering, Bina Nusantara University, Jakarta, Indonesia, 11480. 4th International Conference on Eco Engineering Development 2020.:-**

IOP Conf. Series: Earth and Environmental Science 794 (2021). A major factor affecting food quality is temperature. Food quality can be kept in good condition by storing the raw food materials in the freezer. In this study, ESP32 is utilized to monitor the performance of freezer. The freezer's performance is tracked by taking continuous temperature readings over a predetermined time period, storing the readings, and then computing the results to inform preventive maintenance procedures. By using the Smart temperature monitoring system, Food and Beverage establishments may appropriately perform preventive maintenance, maintain the quality of food, and also boost the staff productivity. The outcome is shown in a consistent measurement of freezer temperature taken throughout the day as a 24-hour format. When the temperature from the DS18B20 is acquired, the programmer running on the controller ESP32 check it against the temperature threshold and send the temperature data through Wi-Fi to the database. If an issue arises, it triggers the buzzer and send a short message service to a phone number set in the system. In order to keep track of temperature, the temperature readings are sent to a cloud database through Wi-Fi. Based on the detained temperature history the user can also anticipate the problems by monitoring the operation of freezer. Maintenance procedure can also be started by the user before the freezer becomes irreparably broken and ceases to work. As an upgradation to the

monitoring system the next action that may be performed additionally is sending of notifications by utilizing various messaging systems such as Email, SMS or Whatsapp in order to improve the monitoring system.

2. “ Smart Refrigerator Using IoT ” Prof.A.A Bamanikar EID, Yogini Nanaware, Nikita Ovhal, Diksha Babar, Pooja Kurawale, Department of Computer, Pune District Education Association’s College of Engineering, Maharashtra, India. International Research Journal of Engineering and Technology (IRJET), Volume: 07. :-

For food storage the most recurrent home and kitchen equipment used ubiquitous is refrigerator. The most illustrious area is kitchen for smart appliances, and the refrigerator is one of those gadgets. This appliance is predominantly used for a diversification of tasks including keeping fruits, vegetables and other items. Any old refrigerator may be transformed into a smart, economical machine employing sensors by installing a smart refrigeration module. The withdrawal of food deterioration, the reduction of sickness, and the creation of better lifestyles for modern individuals are the importance of this study. It is capable of informing people of the food products’ current state using an Android app on the phone. The smart refrigerator module has the capability to forewarn the user and supervise the effectiveness of refrigerator from a distance. The user acquires notifications and information from the programme via an Android app. The smart refrigerator is user-friendly and economically sound.

**3. “ IoT based Interactive Smart Refrigerator ” Murali N.G(1),
Aarthi.S(2), Ethiraj.M(3), Baghavathi Priya.S(4), Rajalakshmi
Engineering College, 602105 Thandalam, Chennai, TamilNadu :-**

The refrigerator is one example of a smart gadget that is increasingly used in daily life as technology advances quickly. By communicating the number of food items accessible to consumers through mobile applications, the proposed Internet of Things (IoT) system identifies the lack of food products. If the weight drops below the threshold, it alerts the users to place an order. With the user's consent, a smart refrigerator system has been developed to detect the weight of food items stored in the refrigerator and make an order for groceries. Therefore, this approach saves time and human labour when forecasting future demands. This technology might be designed to show the contents of the fridge from a distance.

**4. “ IoT for Agriculture : Food Quality and Safety ” Gunawan
Witjaksono(1), Almur Abdelkreem Saeed Rabih(2), Noorhana bt
Yahya(3), Sagir Alva(4), Department of Electrical and Electronic
Engineering, Universiti Teknologi PETRONAS Seri Iskandsr,
Malayasia(1), Department of Fundamental Applied Science, Universiti
Teknologi PETRONAS Seri Iskandar, Malaysia(2), Mechanical
Engineering Department, Mercu Buana University, Indonesia(3).
Published in IOP Conf. Series:-**

Materials Science and Engineering 343. By tracking food conditions and live-sharing the collected data with customers to FSC supervisors, IoT use in the food supply chain (FSC) is thought to improve quality of life. This

essay's goal was to investigate the potential use of IoT in agriculture to trace and monitor the quality and safety of food. Consumer smartphone cameras were used to verify the freshness of food, according to the results of a successful mobile application for food freshness inquiry. This information was distributed to all customers and managers by utilizing IoT technologies. The sensor layer of the IoT technology had a mobile application that demonstrated how the freshness of food can be examined by examining the photo of the food and comparing it to the reference picture. Through the network and application layers of the IoT technology, all customers and food supervisors was informed of the food's status, including whether it is fresh, excellent, or ruined.

5. “ Temperature Control in Domestic Refrigerators and Freezers ”

J.A.Evans(1), A.M.Foster(2), T.Brown(3), Faculty of Engineering, Science and Built Environment, London South Bank University, Langford, Bristol. Published in 3rd IIR International Cold Chain Conference, Twickenham UK, 2014:-

Controlling the temperature of food in the kitchen is crucial. In order to achieve minimal energy consumption, refrigerators are now equipped with thicker insulation and low power compressors. This study compares the findings to those from a survey that was conducted in 1991 on residential refrigerators that are still in use. The report also investigated freezer temperatures in homes.

6. **“ The Heat Exchanger Performance of Shell and Multi Tube Helical Coil as a Heater through the Utilization of a Diesel Machine’s Exhaust Gas ” Zainuddin(1), Jufrizal Nurdin(2), Eswanto Is(3), Mechanical Engineering, Institut Teknologi Medan, Indonesia, Aceh International Journal of Science and Technology. Published on 30, April 2016. :-**

Utilizing a heat exchanger is one method for recovering wasted heat in the industrial setting. The primary goal of the study was to determine if shell and multitube helical coil HE can effectively heat air using diesel engine exhaust as a fuel source. The exhaust thermal gas from the diesel engine was used as an air warmer by the heat exchanger’s shell and multiple tube helical coil. Diesel engine exhaust was used to test the HE-type multi tube helical coil’s capacity to heat the air, and it did so rather effectively. With four different machine rotations, the highest HE efficacy is 67.8%. Inside the drier room the proposed system concerns to maintain the temperature perpetual and can be modified to meet the drying requirements.

7. **“ Smart Refrigerator Using Internet of Things ” Prapulla S.B (1), Dr. Shobha G (2), Dr. Thanuja T.C (3), Department of CSE, RVCE, Bengalurur, India(1), Department of CSE, RVCE, Bengaluru, India(2), VLSI Design and Embedded systems, VTU, Belgaum(3), Journal of Multidisciplinary Engineering Science and Technology, July-2015:-**

Multimedia-capable intelligent gadgets have been making their way into our daily lives. The merchandise was tracked by existing systems using barcode or RFID scanning. The primary purpose of a smart refrigerator is to easily keep track of the food products that might need to be substituted as soon as they run expired. The user is therefore informed each moment the three eggs are consumed. Liquid-sensing containers are monitored

using LDR sensors. The user tells other smart refrigerator users when an order is placed. The smart refrigerator module has the ability to remotely alert the user when the contents of the refrigerator are running low. Additionally, it makes it easier to buy the in-demand food goods online. The notice that is delivered to the user through SMS and email contains a link to the online retailer. With the help of this module, one user may announce a placed order and the remaining individuals can accept it.

Sr. No.	Paper Title and its Author	Details of Publication	Findings
1	WSN based Online Parameter Monitoring in Cold Storage Warehouses in Cloud using IOT concepts	International Research Journal of Engineering and Technology (IRJET) e-ISSN: 2395-0056 Volume: 05 Issue: 07 July-2018	This paper proposes Wireless Sensor Network (WSN) based continuous monitoring of temperature and humidity of the cold storage warehouses using IOT concepts order to ensure that the stored food products are not decayed due to increase in temperature and humidity
2	Cold Storage Traceability System	International Journal for Research in Applied Science & Engineering Technology (IJRASET) 2017	Monitoring of cold storages is a tedious task and it is made easy using Internet of things technology. Continuous monitoring by the real owner of the product can be done in almost real time.

Fig. 2.1. Literature Review Table

CHAPTER 3

METHODOLOGY

CHAPTER No. 3

METHODOLOGY

3.1 **Proposed Work:**

The project's operation is governed by the functionality of the Internet of Things [3]. For this idea to be realized, two architectures are required: the sensing entity in the environment which collects data and the cloud service where those data are stored. Our cold chain monitoring system is based on a network of connected sensors which collect data and upload it to the cloud from which the data is accessible from anywhere. The monitoring system is further divided into two subsystems. One for monitoring the cold storage environment at facilities while the other for monitoring in logistics. The overall system is described in Figure 1 which shows the conceptualization of the two subsystems

The subsystems are based on the NodeMCU prototyping platform. Each of the systems performs these two major tasks: measuring temperature and humidity of the environment and monitoring items moving in and out. On the cloud, the data from facilities follow different paths in reaching the 3 monitoring personnel: the temperature data is uploaded to cloud where it is accessed from the web portal and the inventory information is stored

. Live temperature reading from each wireless sensor node is sent to the web server periodically and details are updated in a central database. The web server is created and an interface is created for the users to track the performance of each appliance in the continuously from anywhere and anytime.

.For this we are using NodeMCU ESP controller which is an open-source electronics platform based on easy-to-use hardware and software with inbuilt wifi module called ESP12. It senses the environment by receiving inputs from sensors. NodeMCU is a microcontroller board based on the ESP12 with 3.3v supply, 8 bit, 4kb of memory and 16 MHz clock frequency. The software coding is done in Arduino programming language and using the provided Arduino Integrated development environment.

The temperature sensor we are using is DS18B20 which is digital sensor and can be programmed on digital pin of the controller.

Block Diagram:-

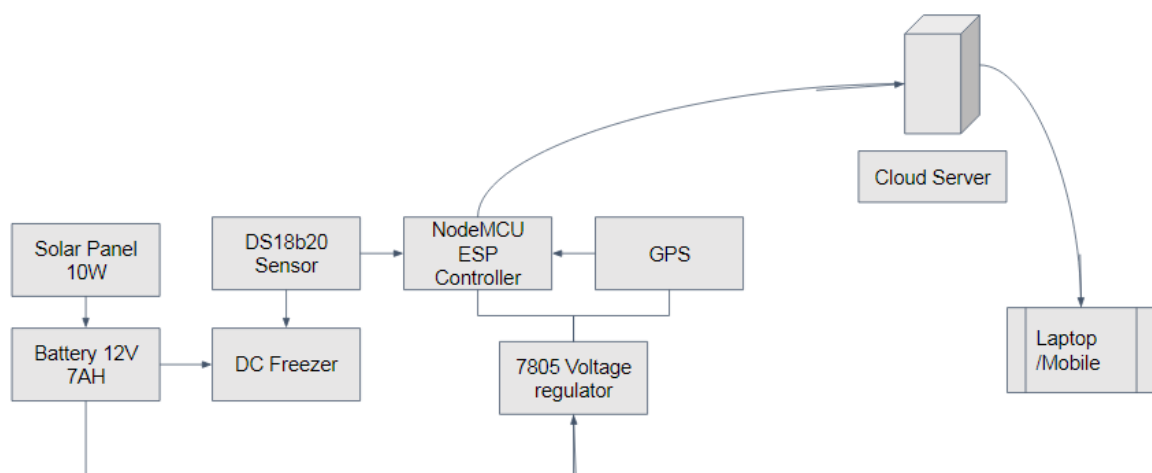


Fig.3.1. BLOCK DIAGRAM

As shown in the diagram, The DS18B20 is Industrial Temperature sensor we are using for this project, The sensor is programmed in Nodemcu to get the temperature of the deep freezer The data will store on cloud server to monitor w.r.t date and time as well as in the form SMS if the When the temperature goes above the set value then SMS will send to the respective contact person with GPS location

3.2 Fault analysis of an IOT-based cold storage monitoring and controlling system:-

Fault analysis of an IoT-based cold storage monitoring and controlling system involves identifying potential issues and failures that may occur in the system. Some of the common faults that may occur in the system include:

- 1. Sensor Failure:** The sensors used in the system may fail due to various reasons such as physical damage, electrical failure, or environmental factors. This can lead to inaccurate readings and affect the overall performance of the system.
- 2. Communication Failure:** The IoT devices used in the system rely on communication networks to transmit data. Any failure in the communication network can lead to loss of data or delayed response time.
- 3. Power Failure:** The system relies on a continuous power supply to function properly. Power outages or voltage fluctuations can cause the system to malfunction or shut down completely.
- 4. Software Failure:** The software used in the system may have bugs or errors that can affect the performance of the system. This can lead to incorrect readings or system failure.
- 5. Human Error:** Human error such as incorrect installation, calibration, or operation of the system can lead to faults in the system.

To address these faults, the system should have built-in redundancy and fail-safe mechanisms to ensure that the system continues to function even in the event of a failure. Regular maintenance and calibration of the system should also be carried out to ensure that the system is functioning optimally. Additionally, the system should have a user-friendly interface that alerts users of any faults or failures in the system, allowing for prompt action to be taken.

3.3 . Proposed Solution:-

A proposed solution for an IoT-based cold storage monitoring and controlling system can include the following components:

1. Sensors: Temperature and humidity sensors will be installed inside the cold storage unit to monitor the conditions inside.

2. Microcontroller: A microcontroller such as Arduino or Raspberry Pi will be used to read the sensor data and send it to the cloud platform.

3. Cloud Platform: A cloud platform such as AWS IoT or Microsoft Azure IoT will be used to store and analyze the data from the sensors.

4. Connectivity: The microcontroller will be connected to the cloud platform using Wi-Fi or cellular connectivity.

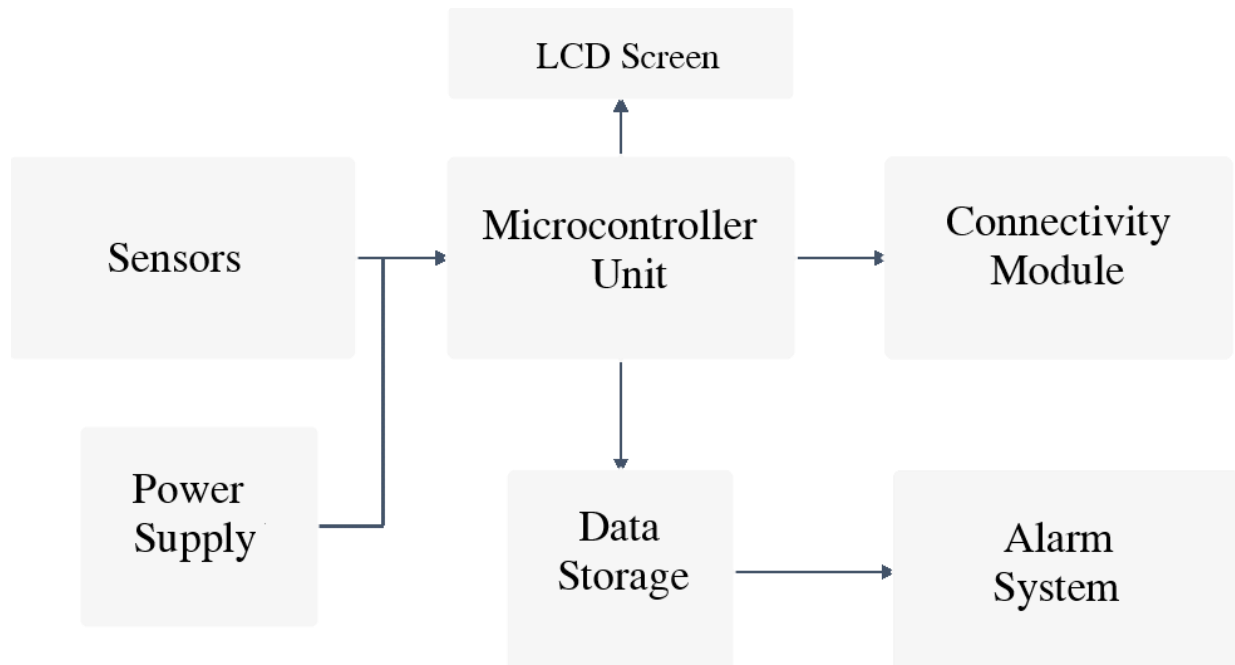
5. User Interface: A web application or mobile application will be developed to provide a user interface for monitoring and controlling the cold storage unit. The user interface will display the current temperature and humidity levels, as well as historical data and trends. The user will also be able to set temperature and humidity thresholds and receive alerts and notifications in case of any fluctuations.

6. Control System: The system will be equipped with a control system that can adjust the temperature and humidity levels automatically based on the user's settings or by analyzing the data from the sensors.

Overall, the proposed solution will provide real-time monitoring and control of the cold storage unit, ensuring that the temperature and humidity levels are always at optimal conditions. This will help to prevent spoilage and wastage of food products, pharmaceuticals, and other perishable items, and improve the overall efficiency and safety of cold storage operation.

3.3 Design of Hardware:-

3.3.1 Basic Block Diagram:-



Designing an IoT-based Cold Storage Monitoring System involves several components and steps. Here's a high-level overview of the hardware design, including the block diagram and circuit diagram elements you would typically need. Please note that creating detailed diagrams requires specific design software and cannot be provided as text, but I can guide you through the components and their connections.

The block diagram for an IoT-based Cold Storage Monitoring System would include the following blocks:

1. Sensors:

- Temperature sensor (e.g., DS18B20)
- Humidity sensor (e.g., DHT22)
- Other environmental sensors as needed (e.g., gas sensors for air quality)

2. Microcontroller Unit (MCU):

- A microcontroller with IoT capabilities (e.g., ESP32, Arduino with Wi-Fi module)

3. Power Supply:

- Power source (e.g., mains power with a voltage regulator or battery)
- Power management circuitry

4. Connectivity Module:

- Wi-Fi module (if not built into the MCU)
- Ethernet module (for wired connectivity)
- Cellular module (for remote areas without Wi-Fi)

5. Data Storage:

- Onboard memory (for short-term data logging)
- Cloud storage service (for long-term data storage and analysis)

6. User Interface:

- Local display (optional, for on-site readings)
- LEDs for status indicators

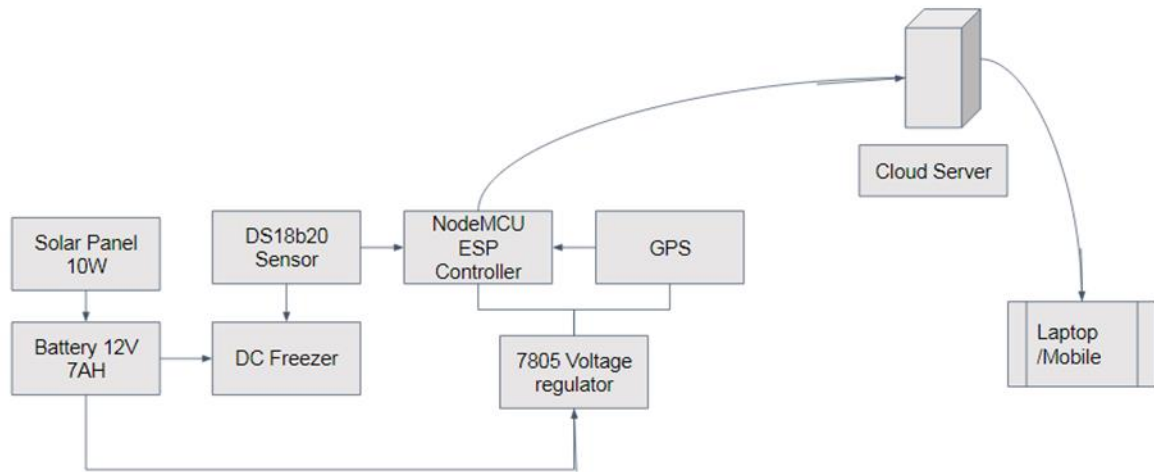
7. Actuators:

- Relays or solid-state switches to control cooling systems

8. Alarm/Notification System:

- Buzzer or alarm system for local alerts
- Push notifications or SMS for remote alerts

3.3.2. Block Diagram connecting to the server process:-



As shown in the diagram, The DS18B20 is Industrial Temperature sensor we are using for this project, The sensor is programmed in Nodemcu to get the temperature of the deep freezer The data will store on cloud server to monitor w.r.t date and time as well as in the form SMS if the When the temperature goes above the set value then SMS will send to the respective contact person with GPS location

3.1.1 Circuit Diagram:-

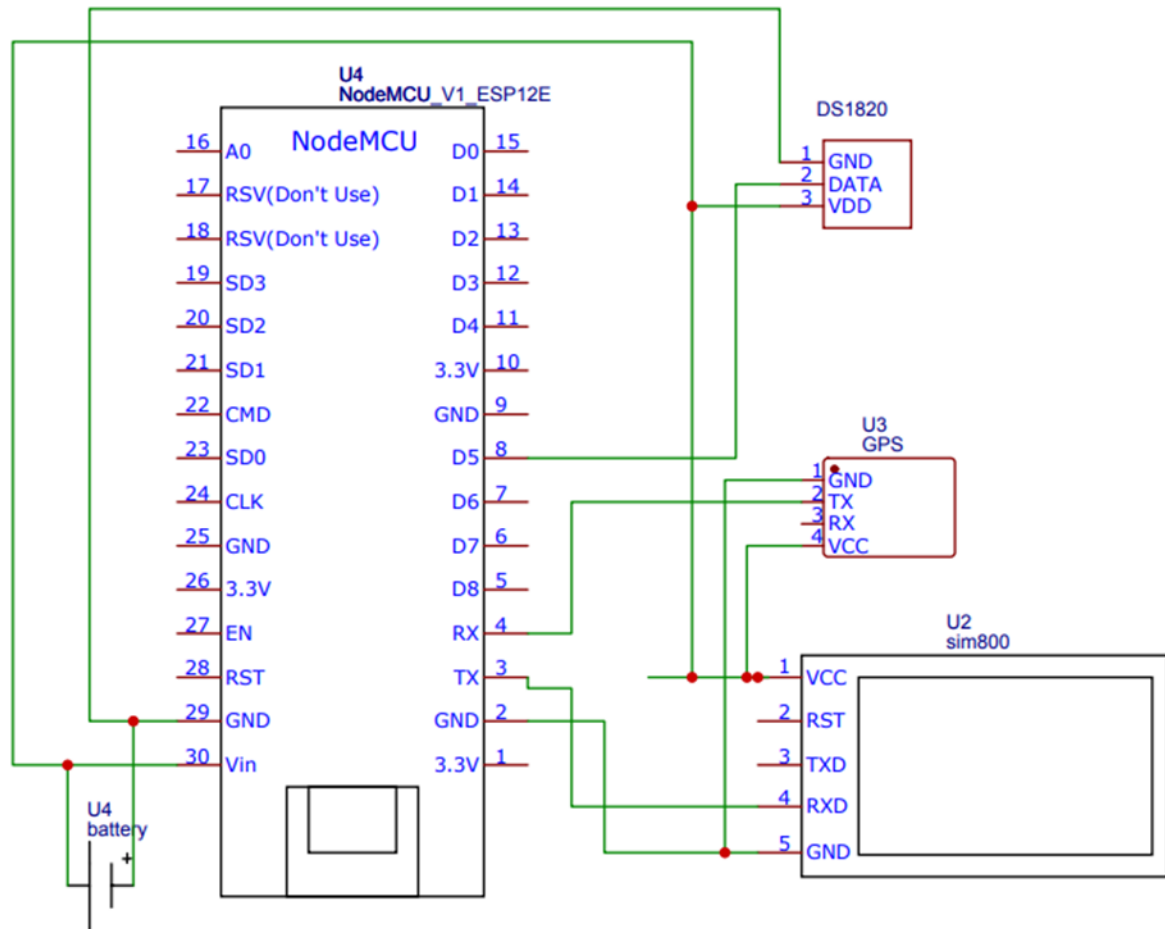


Figure No 3.2(A) Circuit Diagram of the Model

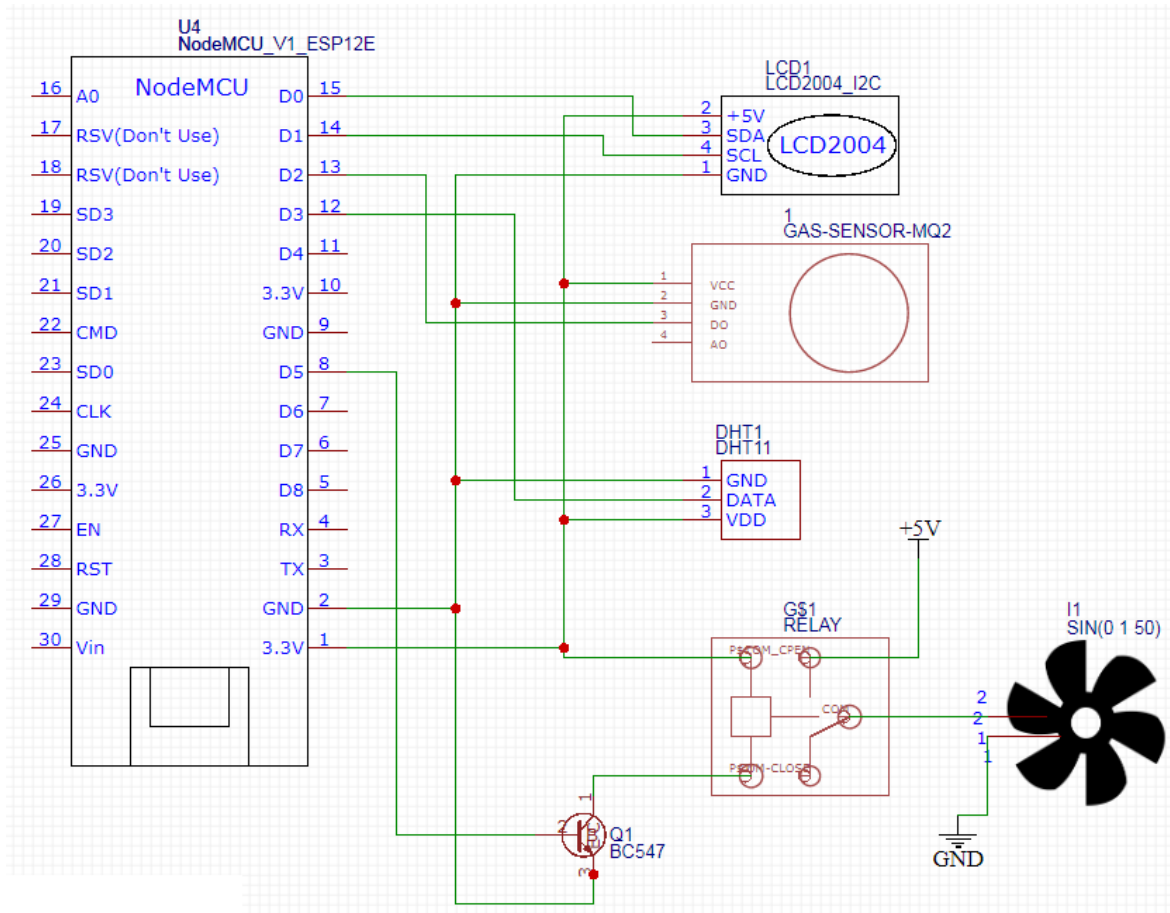


Figure No 3.2(B) Circuit Diagram of the Model

The circuit diagram would show the actual electrical connections between the components. It would typically include:

1. Sensors:

- Each sensor connected to the MCU's GPIO pins with appropriate pull-up/pull-down resistors, if necessary.

2. MCU:

- Power connections, typically 3.3V or 5V, depending on the MCU.
- Ground connections.
- GPIO connections to sensors and actuators.
- UART/I2C/SPI connections if needed for specific modules.

3. Power Supply:

- Voltage regulator to step down mains voltage to the MCU's operating voltage.
- Filtering capacitors to smooth out power supply fluctuations.
- Battery backup system, if included.

4. Connectivity:

- Wi-Fi module connected to the MCU via serial interface (if not integrated).
- Ethernet jack with magnetics, if using wired connectivity.
- SIM card holder and cellular module connections, if using cellular connectivity.

5. Actuators:

- Relay or solid-state switch circuits driven by MCU GPIO pins.
- Protection diodes for inductive loads, if necessary.

6. User Interface:

- Connections for local display (e.g., I2C for an OLED screen).
- LEDs connected with series resistors to MCU GPIO pins.

7. Alarm/Notification System:

- Buzzer connected to an MCU GPIO pin with a transistor driver circuit.

3.4. Construction:-

The construction of an IoT-based cold storage monitoring and controlling system involves the following steps:

1. Sensor Selection: The first step is to select the appropriate sensors for monitoring the temperature and humidity levels of the cold storage unit. There are several types of sensors available in the market, including temperature sensors, humidity sensors, and pressure sensors.

2. Microcontroller Selection: The next step is to select the appropriate microcontroller for processing the data from the sensors and sending it to the cloud. Some of the popular microcontrollers used in IoT applications include Arduino, Raspberry Pi, and ESP32.

3. Cloud Platform Selection: The third step is to select the appropriate cloud platform for storing and analyzing the data from the sensors. Some of the popular cloud platforms used in IoT applications include AWS IoT, Google Cloud IoT, and Microsoft Azure IoT.

4. Connectivity: The next step is to establish connectivity between the microcontroller and the cloud platform. This can be done using various communication protocols such as Wi-Fi, Bluetooth, or cellular.

5. User Interface: The final step is to develop a user interface for monitoring and controlling the cold storage unit. This can be done using a web application or a mobile application that can be accessed from anywhere.

Once the system is constructed, it can be deployed in the cold storage unit, and the data can be monitored and analyzed in real-time. The system can also be configured to send alerts and notifications in case of any fluctuations in temperature or humidity levels, ensuring that the cold storage unit is always operating at optimal conditions.

CHAPTER 4

Components

CHAPTER No. 4

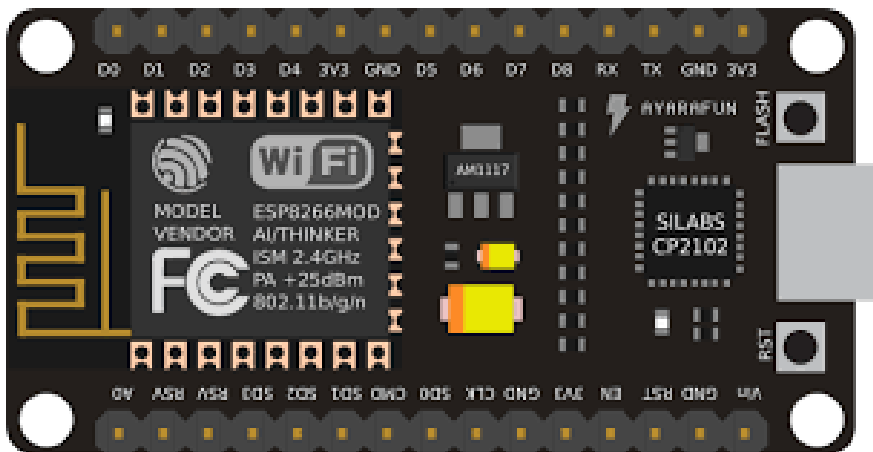
COMPONENTS

4.1 Hardware use:-

4.1.1. Microcontroller ESP8266:-

The board we are using is called “NodeMCUi” and has an ESP8266 module on it, which we will be programming. It comes with the latest version of MicroPython already setup on it, together with all the drivers we are going to use.

The D0, D1, D2, ... numbers printed on the board are different from what Micropython uses – because originally those boards were made for a different software. Make sure to refer to the image below to determine which pins are which.



It has a micro-USB socket for connecting to the computer. On the side is a button for resetting the board. Along the sides of the board are two rows of pins, to which we will be connecting cables.

The symbols meaning is as follows:

3v3 - this is a fancy way to write 3.3V, which is the voltage that the board runs on internally. You can think about this pin like the plus side of a battery.

gnd, G - this is the ground. Think about it like the minus side of the battery.

gpioXX - “gpio” stands for “general purpose input output”. Those are the pins we will be using for sending and receiving signals to and from various devices that we will connect to them. They can act as output – pretty much like a switch that you can connect to plus or to minus with your program. Or they can act as input, telling your program whether they are connected to plus or minus.

a0 - this is the analog pin. It can measure the voltage that is applied to it, but it can only handle up to 3.3V.

5V - this pin is connected with the 5V from your computer. You can also use it to power your board with a battery when it's not connected to the computer. The voltage applied here will be internally converted to the 3.3V that the board needs.

rst - this is a reset button (and a corresponding pin, to which you can connect external button).

Many of the gpio pins have an additional function, we will cover them separately.

4.1.2. Connecting:-

The board you got should already have MicroPython with all the needed libraries flashed on it. In order to access its console, you will need to connect it to your computer with the micro-USB cable, and access the serial interface that appears with a terminal program.

4.1.3. ESP8266:-

Espressif's ESP8266EX delivers highly integrated Wi-Fi SoC solution to meet users' continuous demands for efficient power usage, compact design and reliable performance in the Internet of Things industry. With the complete and self-contained Wi-Fi networking capabilities, ESP8266EX can perform either as a standalone application or as the slave to a host MCU. When ESP8266EX hosts the application, it promptly boots up from the flash. The integrated highspeed cache helps to increase the system performance and optimize the system memory. Also, ESP8266EX can be applied to any microcontroller design as a Wi-Fi adaptor through SPI/SDIO or UART interfaces. ESP8266EX integrates antenna switches, RF balun, power amplifier, low noise receive amplifier, filters and power management modules. The compact design minimizes the PCB size and requires minimal external circuitries. Besides the Wi-Fi functionalities, ESP8266EX also integrates an enhanced version of Tensilica's L106 Diamond series 32-bit processor and on-chip SRAM. It can be interfaced with external sensors and other devices through the GPIOs. Software Development Kit (SDK) provides sample codes for various applications.

Espressif Systems' Smart Connectivity Platform (ESCP) enables sophisticated features including:

- Fast switch between sleep and wakeup mode for energy-efficient purpose;
- Adaptive radio biasing for low-power operation \

- Advance signal processing
- Spur cancellation and RF co-existence mechanisms for common cellular, Bluetooth, DDR, LVDS, LCD interference mitigation

4.1.4 Wi-Fi Key Features:-

- 802.11 b/g/n support
- 802.11n support (2.4 GHz), up to 72.2 Mbps
- Defragmentation
- 2 x virtual Wi-Fi interface
- Automatic beacon monitoring (hardware TSF)
- Support Infrastructure BSS Station mode/SoftAP mode/Promiscuous mode •
- Antenna diversity

4.1.5. Applications

- Home appliances
 - Home automation
- Smart plugs and lights
 - Industrial wireless control
- Baby monitors
- IP cameras
- Sensor networks
- Wearable electronics
- Wi-Fi location-aware devices
- Security ID tags
- Wi-Fi position system beacons

4.1.6 CPU :-

The ESP8266EX integrates a Tensilica L106 32-bit RISC processor, which achieves extralow power consumption and reaches a maximum clock speed of 160 MHz. The Real-Time Operating System (RTOS) and Wi-Fi stack allow 80% of the processing power to be available for user application programming and development.

The CPU includes the interfaces as below:

- Programmable RAM/ROM interfaces (iBus), which can be connected with memory controller, and can also be used to visit flash.
- Data RAM interface (dBus), which can connected with memory controller.
- AHB interface which can be used to visit the register.

Memory ESP8266EX Wi-Fi SoC integrates memory controller and memory units including SRAM and ROM. MCU can access the memory units through iBus, dBus, and AHB interfaces. All memory units can be accessed upon request, while a memory arbiter will decide the running sequence according to the time when these requests are received by the processor. According to our current version of SDK, SRAM space available to users is assigned as below.

- RAM size < 50 kB, that is, when ESP8266EX is working under the Station mode and connects to the router, the maximum programmable space accessible in Heap + Data section is around 50 kB.
- There is no programmable ROM in the SoC. Therefore, user program must be stored in an external SPI flash.

External Flash ESP8266EX uses external SPI flash to store user programs, and supports up to 16 MB memory capacity theoretically.

The minimum flash memory of ESP8266EX is shown below:

- OTA disabled: 512 kB at least
- OTA enabled: 1 MB at least

High Frequency Clock The high frequency clock on ESP8266EX is used to drive both transmit and receive mixers. This clock is generated from internal crystal oscillator and external crystal. The crystal frequency ranges from 24 MHz to 52 MHz. The internal calibration inside the crystal oscillator ensures that a wide range of crystals can be used, nevertheless the quality of the crystal is still a factor to consider to have reasonable phase noise and good Wi-Fi sensitivity. Refer to Table 3-1 to measure the frequency offset

Radio ESP8266EX radio consists of the following blocks.

- 2.4 GHz receiver
- 2.4 GHz transmitter
 - High speed clock generators and crystal oscillator
- Bias and regulators
- Power management

➤ **2.4 GHz Receiver :-**

The 2.4 GHz receiver down-converts the RF signals to quadrature baseband signals and converts them to the digital domain with 2 high resolution high speed ADCs. To adapt to varying signal channel conditions, RF filters, automatic gain control (AGC), DC offset cancelation circuits and baseband filters are integrated within ESP8266EX.

➤ **2.4 GHz Transmitter :-**

The 2.4 GHz transmitter up-converts the quadrature baseband signals to 2.4 GHz, and drives the antenna with a high-power CMOS power amplifier. The function of digital calibration further improves the linearity of the power amplifier, enabling a state of art performance of delivering +19.5 dBm average TX power for 802.11b transmission and +18 dBm for 802.11n (MSC0) transmission.

Additional calibrations are integrated to offset any imperfections of the radio, such as:

- Carrier leakage

- I/Q phase matching
- Baseband nonlinearities These built-in calibration functions reduce the product test time and make the test equipment unnecessary.

4.2. Connecting Hardware:-

4.2.1 Clock Generator :-

The clock generator generates quadrature 2.4 GHz clock signals for the receiver and transmitter. All components of the clock generator are integrated on the chip, including all inductors, varactors, loop filters, linear voltage regulators and dividers. The clock generator has built-in calibration and self test circuits. Quadrature clock phases and phase noise are optimized on-chip with patented calibration algorithms to ensure the best performance of the receiver and transmitter.

Wi-Fi ESP8266EX implements TCP/IP and full 802.11 b/g/n WLAN MAC protocol. It supports Basic Service Set (BSS) STA and SoftAP operations under the Distributed Control Function (DCF). Power management is handled with minimum host interaction to minimize activeduty period.

4.2.2 Wi-Fi Radio and Baseband :-

The ESP8266EX Wi-Fi Radio and Baseband support the following features:

- 802.11b and 802.11g
- 802.11n MCS0-7 in 20 MHz bandwidth
- 802.11n 0.4 μ s guard-interval
- up to 72.2 Mbps of data rate

4.2.3 Wi-Fi MAC

The ESP8266EX Wi-Fi MAC applies low-level protocol functions automatically, as follows:

- 2 × virtual Wi-Fi interfaces
- Infrastructure BSS Station mode/SoftAP mode/Promiscuous mode
- Request To Send (RTS), Clear To Send (CTS) and Immediate Block ACK
- Defragmentation
- CCMP (CBC-MAC, counter mode), TKIP (MIC, RC4), WEP (RC4) and CRC
 - Automatic beacon monitoring (hardware TSF)
- Dual and single antenna Bluetooth co-existence support with optional simultaneous receive (Wi-Fi/Bluetooth) capability

4.2.4. Power Management :-

ESP8266EX is designed with advanced power management technologies and intended for mobile devices, wearable electronics and the Internet of Things applications. The low-power architecture operates in the following modes:

- Active mode: The chip radio is powered on. The chip can receive, transmit, or listen.
- Modem-sleep mode: The CPU is operational. The Wi-Fi and radio are disabled.
- Light-sleep mode: The CPU and all peripherals are paused. Any wake-up events (MAC, host, RTC timer, or external interrupts) will wake up the chip.
- Deep-sleep mode: Only the RTC is operational and all other part of the chip are powered off

General Purpose Input/Output Interface (GPIO) ESP8266EX has 17 GPIO pins which can be assigned to various functions by programming the appropriate registers. Each GPIO PAD can be configured with internal pull-up or pull-down (XPD_DCDC can only be configured with internal pull-down, other GPIO PAD can only be configured with internal pull-up), or set to high impedance. When configured as an input, the data are stored in software registers; the input can also be set to edge-trigger or level trigger CPU interrupts. In short, the IO pads are bi-directional, non-inverting and tristate, which includes input and output buffer with tristate control inputs. These pins, when working as GPIOs, can be multiplexed

with other functions such as I2C, I2S, UART, PWM, and IR Remote Control, etc. For low power operations, the GPIOs can also be set to hold their state. For instance, when the IOs are not driven by internal and external circuits, all outputs will hold their states before the chip entered the low power modes. The required drive strength is small— 5 μ A or more is enough to pull apart the latch.

Secure Digital Input/Output Interface (SDIO) ESP8266EX has one Slave SDIO, the definitions of which are described as Table 4-1, which supports 25 MHz SDIO v1.1 and 50 MHz SDIO v2.0, and 1 bit/4 bit SD mode and SPI mode

4.2.5. DS18B20 Temperature Sensor:-

The DS18B20 digital thermometer provides 9-bit to 12-bit Celsius temperature measurements and has an alarm function with nonvolatile user-programmable upper and lower trigger points. The DS18B20 communicates over a 1-Wire bus that by definition requires only one data line (and ground) for communication with a central microprocessor. In addition, the DS18B20 can derive power directly from the data line (“parasite power”), eliminating the need for an external power supply. Each DS18B20 has a unique 64-bit serial code, which allows multiple DS18B20s to function on the same 1-Wire bus. Thus, it is simple to use one microprocessor to control many DS18B20s distributed over a large area. Applications that can benefit from this feature include HVAC environmental controls, temperature monitoring systems inside buildings, equipment, or machinery, and process monitoring and control systems.

Applications

- Thermostatic Controls
- Industrial Systems
- Consumer Products
- Thermometers
- Thermally Sensitive Systems

Benefits and Features

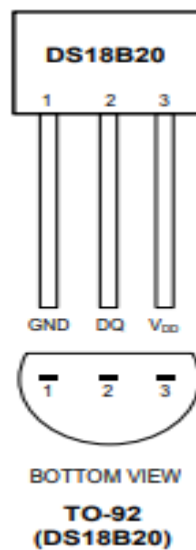
- Unique 1-Wire® Interface Requires Only One Port

Pin for Communication

- Reduce Component Count with Integrated

Temperature Sensor and EEPROM

- Measures Temperatures from -55°C to +125°C (-67°F to +257°F)
- $\pm 0.5^\circ\text{C}$ Accuracy from -10°C to +85°C
- Programmable Resolution from 9 Bits to 12 Bits
- No External Components Required
- Parasitic Power Mode Requires Only 2 Pins for Operation (DQ and GND)
- Simplifies Distributed Temperature-Sensing Applications with Multidrop Capability
- Each Device Has a Unique 64-Bit Serial Code Stored in On-Board ROM
- Flexible User-Definable Nonvolatile (NV) Alarm Settings with Alarm Search Command Identifies Devices with Temperatures Outside Programmed Limits
- Available in 8-Pin SO (150 mils), 8-Pin μSOP , and 3-Pin TO-92 Packages



Overview Figure 3 shows a block diagram of the DS18B20, and pin descriptions are given in the Pin Description table. The 64-bit ROM stores the device's unique serial code. The scratchpad memory contains the 2-byte temperature register that stores the digital output from the temperature sensor. In addition, the scratchpad provides access to the 1-byte upper and lower alarm trigger registers (TH and TL) and the 1-byte configuration register. The configuration register allows the user to set the resolution of the temperature-to-digital conversion to 9, 10, 11, or 12 bits. The TH, TL, and configuration registers are nonvolatile (EEPROM), so they will retain data when the device is powered down. The DS18B20 uses Maxim's exclusive 1-Wire bus protocol that implements bus communication using one control signal. The control line requires a weak pullup resistor since all devices are linked to the bus via a 3-state or open-drain port (the DQ pin in the case of the DS18B20). In this bus system, the microprocessor (the master device) identifies and addresses devices on the bus using each device's unique 64-bit code. Because each device has a unique code, the number of devices that can be addressed on one bus is virtually unlimited. The 1-Wire bus protocol, including detailed explanations of the commands and "time slots," is covered in the 1-Wire Bus System section. Another feature of the DS18B20 is the ability to operate without an external power supply. Power is instead supplied through the 1-Wire pullup

resistor through the

DQ pin when the bus is high. The high bus signal also charges an internal capacitor (CPP), which then supplies power to the device when the bus is low. This method of deriving power from the 1-Wire bus is referred to as “parasite power.” As an alternative, the DS18B20 may also be powered by an external supply on VDD.

Operation—Measuring Temperature

The core functionality of the DS18B20 is its direct-digital temperature sensor. The resolution of the temperature sensor is user-configurable to 9, 10, 11, or 12 bits, corresponding to increments of 0.5°C, 0.25°C, 0.125°C, and 0.0625°C, respectively. The default resolution at power-up is 12-bit. The DS18B20 powers up in a lowpower idle state. To initiate a temperature measurement and A-to-D conversion, the master must issue a Convert T [44h] command. Following the conversion, the resulting thermal data is stored in the 2-byte temperature register in the scratchpad memory and the DS18B20 returns to its idle state. If the DS18B20 is powered by an external supply, the master can issue “read time slots” (see the 1-Wire Bus System section) after the Convert T command and the DS18B20 will respond by transmitting 0 while the temperature conversion is in progress and 1 when the conversion is done. If the DS18B20 is powered with parasite power, this notification technique cannot be used since the bus must be pulled high by a strong pullup during the entire temperature conversion. The bus requirements for parasite power are explained in detail in the Powering the DS18B20 section.

The DS18B20 output temperature data is calibrated in degrees Celsius; for Fahrenheit applications, a lookup table or conversion routine must be used. The temperature data is stored as a 16-bit sign-extended two’s complement number in the temperature register (see Figure 4). The sign bits (S) indicate if the temperature is positive or negative: for positive numbers $S = 0$ and for negative numbers $S = 1$. If the DS18B20 is configured for 12-bit resolution, all bits in the temperature register will contain valid data. For 11-bit resolution, bit 0 is undefined. For 10-bit resolution, bits 1 and 0 are undefined, and for 9-bit resolution bits 2, 1, and 0 are

undefined. Table 1 gives examples of digital output data and the corresponding temperature reading for 12-bit resolution conversions.

Operation—Alarm Signaling After the DS18B20 performs a temperature conversion, the temperature value is compared to the user-defined two's complement alarm trigger values stored in the 1-byte TH and TL registers (see Figure 5). The sign bit (S) indicates if the value is positive or negative: for positive numbers $S = 0$ and for negative numbers $S = 1$. The TH and TL registers are nonvolatile (EEPROM) so they will retain data when the device is powered down. TH and TL can be accessed through bytes 2 and 3 of the scratchpad as explained in the Memory section. Only bits 11 through 4 of the temperature register are used in the TH and TL comparison since TH and TL are 8-bit registers. If the measured temperature is lower than

Powering the DS18B20 The DS18B20 can be powered by an external supply on the VDD pin, or it can operate in “parasite power” mode, which allows the DS18B20 to function without a local external supply. Parasite power is very useful for applications that require remote temperature sensing or that are very space constrained. Figure 3 shows the DS18B20's parasite-power control circuitry, which “steals” power from the 1-Wire bus via the DQ pin when the bus is high. The stolen charge powers the DS18B20 while the bus is high, and some of the charge is stored on the parasite power capacitor (CPP) to provide power when the bus is low. When the DS18B20 is used in parasite power mode, the VDD pin must be connected to ground. In parasite power mode, the 1-Wire bus and CPP can provide sufficient current to the DS18B20 for most operations as long as the specified timing and voltage requirements are met (see the DC Electrical Characteristics and AC Electrical Characteristics). However, when the DS18B20 is performing temperature conversions or copying data from the scratchpad memory to EEPROM, the operating current can be as high as 1.5mA. This current can cause an unacceptable voltage drop across the weak 1-Wire pullup resistor and is more current than can be supplied by CPP. To assure that the DS18B20 has

sufficient supply current, it is necessary to provide a strong pullup on the 1-Wire bus whenever temperature conversions are taking place or data is being copied from the scratchpad to EEPROM. This can be accomplished by using a MOSFET to pull the bus directly to the rail as shown in Figure 6. The 1-Wire bus must be switched to the strong pullup within 10 μ s (max) after a Convert T [44h] or Copy Scratchpad [48h] command is issued, and the bus must be held high by the pullup for the duration of the conversion (t_{CONV}) or data transfer ($t_{WR} = 10\text{ms}$). No other activity can take place on the 1-Wire bus while the pullup is enabled. The DS18B20 can also be powered by the conventional method of connecting an external power supply to the VDD pin, as shown in Figure 7. The advantage of this method is that the MOSFET pullup is not required, and the 1-Wire bus is free to carry other traffic during the temperature conversion time. The use of parasite power is not recommended for temperatures above +100°C since the DS18B20 may not be able to sustain communications due to the higher leakage currents that can exist at these temperatures. For applications in which such temperatures are likely, it is strongly recommended that the DS18B20 be powered by an external power supply. In some situations the bus master may not know whether the DS18B20s on the bus are parasite powered or powered by external supplies. The master needs this information to determine if the strong bus pullup should be used during temperature conversions. To get this information, the master can issue a Skip ROM [CCh] command followed by a Read Power Supply [B4h] command followed by a “read time slot”. During the read time slot, parasite powered DS18B20s will pull the bus low, and externally powered DS18B20s will let the bus remain high. If the bus is pulled low, the master knows that it must supply the strong pullup on the 1-Wire bus during temperature conversions

64-BIT Lasered ROM code Each DS18B20 contains a unique 64-bit code (see Figure 8) stored in ROM. The least significant 8 bits of the ROM code contain the DS18B20's 1-Wire family code: 28h. The next 48 bits contain a unique serial number. The most significant 8 bits contain a cyclic redundancy check (CRC) byte

that is calculated from the first 56 bits of the ROM code. A detailed explanation of the CRC bits is provided in the CRC Generation section. The 64-bit ROM code and associated ROM function control logic allow the DS18B20 to operate as a 1-Wire device using the protocol detailed in the 1-Wire Bus System section.

Memory The DS18B20's memory is organized as shown in Figure 9. The memory consists of an SRAM scratchpad with nonvolatile EEPROM storage for the high and low alarm trigger registers (TH and TL) and configuration register. Note that if the DS18B20 alarm function is not used, the TH and TL registers can serve as general-purpose memory. All memory commands are described in detail in the DS18B20 Function Commands section. Byte 0 and byte 1 of the scratchpad contain the LSB and the MSB of the temperature register, respectively. These bytes are read-only. Bytes 2 and 3 provide access to TH and TL registers. Byte 4 contains the configuration register data, which is explained in detail in the Configuration Register section. Bytes 5, 6, and 7 are reserved for internal use by the device and cannot be overwritten. Byte 8 of the scratchpad is read-only and contains the CRC code for bytes 0 through 7 of the scratchpad. The DS18B20 generates this CRC using the method described in the CRC Generation section. Data is written to bytes 2, 3, and 4 of the scratchpad using the Write Scratchpad [4Eh] command; the data must be transmitted to the DS18B20 starting with the least significant bit of byte 2. To verify data integrity, the scratchpad can be read (using the Read Scratchpad [BEh] command) after the data is written. When reading the scratchpad, data is transferred over the 1-Wire bus starting with the least significant bit of byte 0. To transfer the TH, TL and configuration data from the scratchpad to EEPROM, the master must issue the Copy Scratchpad [48h] command. Data in the EEPROM registers is retained when the device is powered down; at power-up the EEPROM data is reloaded into the corresponding scratchpad locations. Data can also be reloaded from EEPROM to the scratchpad at any time using the Recall E2 [B8h] command. The master can issue read time slots following the Recall E2 command and the DS18B20 will indicate the status

of the recall by transmitting 0 while the recall is in progress and 1 when the recall is done.

Configuration Register Byte 4 of the scratchpad memory contains the configuration register, which is organized as illustrated in Figure 10. The user can set the conversion resolution of the DS18B20 using the R0 and R1 bits in this register as shown in Table 2. The power-up default of these bits is R0 = 1 and R1 = 1 (12-bit resolution). Note that there is a direct tradeoff between resolution and conversion time. Bit 7 and bits 0 to 4 in the configuration register are reserved for internal use by the device and cannot be overwritten. CRC Generation CRC bytes are provided as part of the DS18B20's 64-bit ROM code and in the 9th byte of the scratchpad memory. The ROM code CRC is calculated from the first 56 bits of the ROM code and is contained in the most significant byte of the ROM. The scratchpad CRC is calculated from the data stored in the scratchpad, and therefore it changes when the data in the scratchpad changes. The CRCs provide the bus master with a method of data validation when data is read from the DS18B20. To verify that data has been read correctly, the bus master must re-calculate the CRC from the received data and then compare this value to either the ROM code CRC (for ROM reads) or to the scratchpad CRC (for scratchpad reads). If the calculated CRC matches the read CRC, the data has been received error free. The comparison of CRC values and the decision to continue with an operation are determined entirely by the bus master. There is no circuitry inside the DS18B20 that prevents a command sequence from proceeding if the DS18B20 CRC (ROM or scratchpad) does not match the value generated by the bus master. The equivalent polynomial function of the CRC (ROM or scratchpad) is: $CRC = X^8 + X^5 + X^4 + 1$ The bus master can re-calculate the CRC and compare it to the CRC values from the DS18B20 using the polynomial generator shown in Figure 11. This circuit consists of a shift register and XOR gates, and the shift register bits are initialized to 0. Starting with the least significant bit of the ROM code or the least significant bit of byte 0 in the scratchpad, one bit at a time should be shifted into the shift register.

After shifting in the 56th bit from the ROM or the most significant bit of byte 7 from the scratchpad, the polynomial generator will contain the recalculated CRC. Next, the 8-bit ROM code or scratchpad CRC from the DS18B20 must be shifted into the circuit. At this point, if the re-calculated CRC was correct, the shift register will contain all 0s. Additional information about the Maxim 1-Wire cyclic redundancy check is available in Application Note 27: Understanding and Using Cyclic Redundancy Checks with Maxim iButton Products.

1-Wire Bus System The 1-Wire bus system uses a single bus master to control one or more slave devices. The DS18B20 is always a slave. When there is only one slave on the bus, the system is referred to as a “single-drop” system; the system is “multidrop” if there are multiple slaves on the bus. All data and commands are transmitted least significant bit first over the 1-Wire bus. The following discussion of the 1-Wire bus system is broken down into three topics: hardware configuration, transaction sequence, and 1-Wire signaling (signal types and timing).

Hardware Configuration The 1-Wire bus has by definition only a single data line. Each device (master or slave) interfaces to the data line via an open-drain or 3-state port. This allows each device to “release” the data line when the device is not transmitting data so the bus is available for use by another device. The 1-Wire port of the DS18B20 (the DQ pin) is open drain with an internal circuit equivalent to that shown in Figure 12. The 1-Wire bus requires an external pullup resistor of approximately 5k Ω ; thus, the idle state for the 1-Wire bus is high. If for any reason a transaction needs to be suspended, the bus MUST be left in the idle state if the transaction is to resume. Infinite recovery time can occur between bits so long as the 1-Wire bus is in the inactive (high) state during the recovery period. If the bus is held low for more than 480 μ s, all components on the bus will be reset.

Transaction Sequence The transaction sequence for accessing the DS18B20 is as follows: Step 1. Initialization Step 2. ROM Command (followed by any required data exchange) Step 3. DS18B20 Function Command (followed by any required data exchange) It is very important to follow this sequence every time

the DS18B20 is accessed, as the DS18B20 will not respond if any steps in the sequence are missing or out of order. Exceptions to this rule are the Search ROM [F0h] and Alarm Search [ECh] commands. After issuing either of these ROM commands, the master must return to Step 1 in the sequence.

Initialization All transactions on the 1-Wire bus begin with an initialization sequence. The initialization sequence consists of a reset pulse transmitted by the bus master followed by presence pulse(s) transmitted by the slave(s). The presence pulse lets the bus master know that slave devices (such as the DS18B20) are on the bus and are ready to operate. Timing for the reset and presence pulses is detailed in the 1-Wire Signaling section.

ROM Commands After the bus master has detected a presence pulse, it can issue a ROM command. These commands operate on the unique 64-bit ROM codes of each slave device and allow the master to single out a specific device if many are present on the 1-Wire bus. These commands also allow the master to determine how many and what types of devices are present on the bus or if any device has experienced an alarm condition. There are five ROM commands, and each command is 8 bits long. The master device must issue an appropriate ROM command before issuing a DS18B20 function command. A flowchart for operation of the ROM commands is shown in Figure 13.

Search Rom [F0h] When a system is initially powered up, the master must identify the ROM codes of all slave devices on the bus, which allows the master to determine the number of slaves and their device types. The master learns the ROM codes through a process of elimination that requires the master to perform a Search ROM cycle (i.e., Search ROM command followed by data exchange) as many times as necessary to identify all of the slave devices.

1-Wire Signaling The DS18B20 uses a strict 1-Wire communication protocol to ensure data integrity. Several signal types are defined by this protocol: reset pulse, presence pulse, write 0, write 1, read 0, and read 1. The bus master initiates all these signals, with the exception of the presence pulse.

Initialization Procedure—Reset And Presence Pulses All communication with the DS18B20 begins with an

initialization sequence that consists of a reset pulse from the master followed by a presence pulse from the DS18B20. This is illustrated in Figure 15. When the DS18B20 sends the presence pulse in response to the reset, it is indicating to the master that it is on the bus and ready to operate. During the initialization sequence the bus master transmits (TX) the reset pulse by pulling the 1-Wire bus low for a minimum of 480 μ s. The bus master then releases the bus and goes into receive mode (RX). When the bus is released, the 5k Ω pullup resistor pulls the 1-Wire bus high. When the DS18B20 detects this rising edge, it waits 15 μ s to 60 μ s and then transmits a presence pulse by pulling the 1-Wire bus low for 60 μ s to 240 μ s.

Read/Write Time Slots The bus master writes data to the DS18B20 during write time slots and reads data from the DS18B20 during read time slots. One bit of data is transmitted over the 1-Wire bus per time slot.

Write Time Slots There are two types of write time slots: “Write 1” time slots and “Write 0” time slots. The bus master uses a Write 1 time slot to write a logic 1 to the DS18B20 and a Write 0 time slot to write a logic 0 to the DS18B20. All write time slots must be a minimum of 60 μ s in duration with a minimum of a 1 μ s recovery time between individual write slots. Both types of write time slots are initiated by the master pulling the 1-Wire bus low. To generate a Write 1 time slot, after pulling the 1-Wire bus low, the bus master must release the 1-Wire bus within 15 μ s. When the bus is released, the 5k Ω pullup resistor will pull the bus high. To generate a Write 0 time slot, after pulling the 1-Wire bus low, the bus master must continue to hold the bus low for the duration of the time slot (at least 60 μ s). The DS18B20 samples the 1-Wire bus during a window that lasts from 15 μ s to 60 μ s after the master initiates the write time slot. If the bus is high during the sampling window, a 1 is written to the DS18B20. If the line is low, a 0 is written to the DS18B20.

Absolute Maximum Ratings

Voltage Range on Any Pin Relative to Ground....-0.5V to +6.0V Operating Temperature Range..... -55°C to +125°C

4.2.6. GSM SIM 800L:-

Mini GSM / GPRS breakout board is based on SIM800L module, supports quad-band GSM/GPRS network, available for GPRS and SMS message data remote transmission.

The board features compact size and low current consumption. With power saving technique, the current consumption is as low as 1mA in sleep mode.

It communicates with microcontroller via UART port, supports command including 3GPP TS 27.007, 27.005 and SIMCOM enhanced AT Commands.

Features:-

Quad-band 850/900/1800/1900MHz

Connect onto any global GSM network with any 2G SIM (in the USA, T-Mobile is suggested)

Make and receive voice calls using a headset or an external 8 speaker and electret microphone

Send and receive SMS messages

Send and receive GPRS data (TCP/IP, HTTP, etc.)

Scan and receive FM radio broadcasts

Lead out buzzer and vibrational motor control port

AT command interface with "auto baud" detection

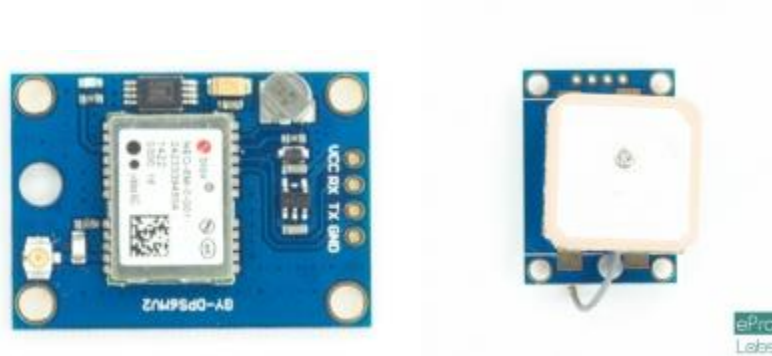


Figure No 4.2.6. GSM SIM 800L

4.2.7 GPS:-

The NEO-6 module series is a family of stand-alone GPS receivers featuring the high performance u-blox 6 positioning engine. These flexible and cost effective receivers offer numerous connectivity options in a miniature 16 x 12.2 x 2.4 mm package. Their compact architecture and power and memory options make NEO-6 modules ideal for battery operated mobile devices with very strict cost and space constraints. The 50-channel u-blox 6 positioning engine boasts a Time-To-First-Fix (TTFF) of under 1 second. The dedicated acquisition engine, with 2 million correlators, is capable of massive parallel time/frequency space searches, enabling it to find satellites instantly. Innovative design and technology suppresses jamming

sources and mitigates multipath effects, giving NEO-6 GPS receivers excellent navigation performance even in the most challenging environments.



The NEO-6M GPS module is a well-performing complete GPS receiver with a built-in 25 x 25 x 4mm ceramic antenna, which provides a strong satellite search capability. With the power and signal indicators, you can monitor the status of the module. Thanks to the data backup battery, the module can save the data when the main power is shut down accidentally. Its 3mm mounting holes can ensure easy assembly on your aircraft, which thus can fly steadily at a fixed position, return to Home automatically, and automatic waypoint flying, etc. Or you can apply it on your smart robot car for automatic returning or heading to a certain destination, making it a real "smart" bot!

Features:-

- 1) A complete GPS module with an active antenna integrated, and a built-in EEPROM to save configuration parameter data.
- 2) Built-in 25 x 25 x 4mm ceramic active antenna provides strong satellite search capability.
- 3) Equipped with power and signal indicator lights and data backup battery.
- 4) Power supply: 3-5V; Default baud rate: 9600bps.
- 5) Interface: RS232 TTL

Test:-

In this test, we will send the positioning data collected by the NEO-6M GPS Module to the software on the PC, and compare this result with that of a standard GPS device. Thus we can know whether this module works or not.

Preparations

1 x USB-to-Serial Module (e.g. FTDI Module)

1 x USB Cable

Laptop

1 x NEO-6M GPS Module

CHAPTER 5

CHAPTER 5

5.1. SOFTWARE USE :-

The hardware programming is done on Arduino Compiler software. The flow of data is passed through the layers built upon of TCP/IP stack following a machine-to-machine (M2M) or Internet of Things connectivity protocol. The live data from the sensors through the Wi-Fi module is sent over to the database for storage. The Server is built so as to monitor the data and visualize these load graphs online. The server is Thingspeak or cloud server with MySQL database and web server to show our website called the Apache HTTP Server, The MySQL relational database management system (RDBMS), and The PHP programming language. The flow of data to the web server, its connection to the database etc is done using the php scripts. One php script is responsible for connecting to the database. It contains the configuration settings for the database. If we want to see the data from the database all we have to do is to visit the table file from a browser. The php script will connect to database and it will ask the database to return all the stored data. Then it will display all the data in an HTML table. The data analytics is done so as to compare and notify the users that which appliances are consuming the maximum power and which has max temperature.

5.5.1 CLOUD SERVER:-

Cloud storage is a model of computer data storage in which the digital data is stored in logical pools. The physical storage spans multiple servers (sometimes in multiple locations), and the physical environment is typically owned and managed by a hosting company. These cloud storage providers are responsible for keeping the data available and accessible, and the physical environment protected and running. People and organizations buy or lease storage capacity from the providers to store user, organization, or application data.

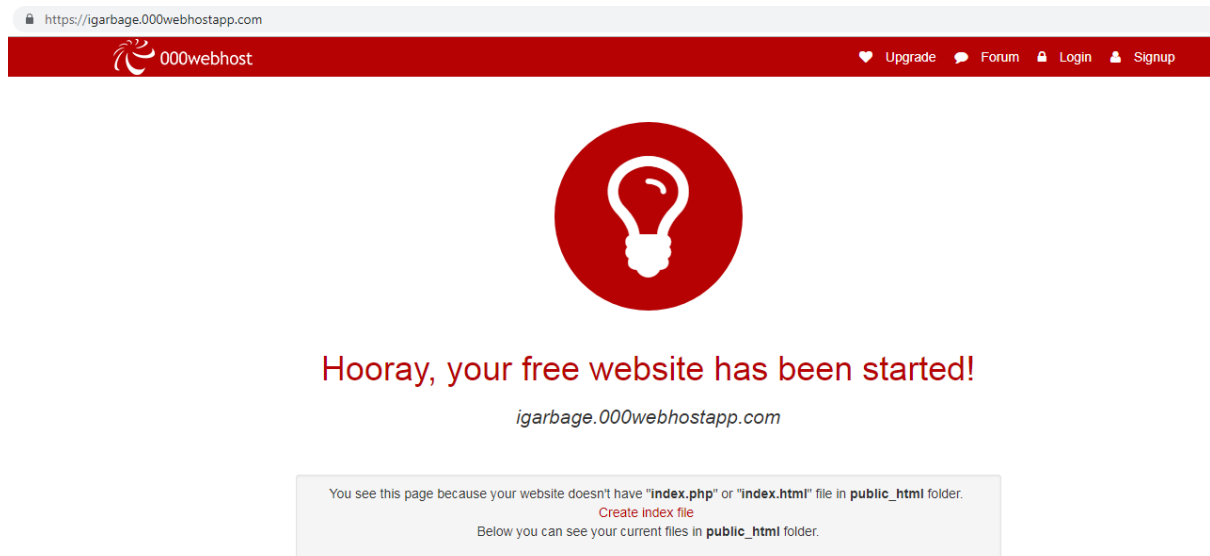
Cloud storage services may be accessed through a colocated cloud computing service, a web service application programming interface (API) or by applications that utilize the API, such as cloud desktop storage, a cloud storage gateway or Web-based content management systems.

Cloud storage is based on highly virtualized infrastructure and is like broader cloud computing in terms of accessible interfaces, near-instant elasticity and scalability, multi-tenancy, and metered resources. Cloud storage services can be utilized from an off-premises service (Amazon S3) or deployed on-premises (ViON Capacity Services).

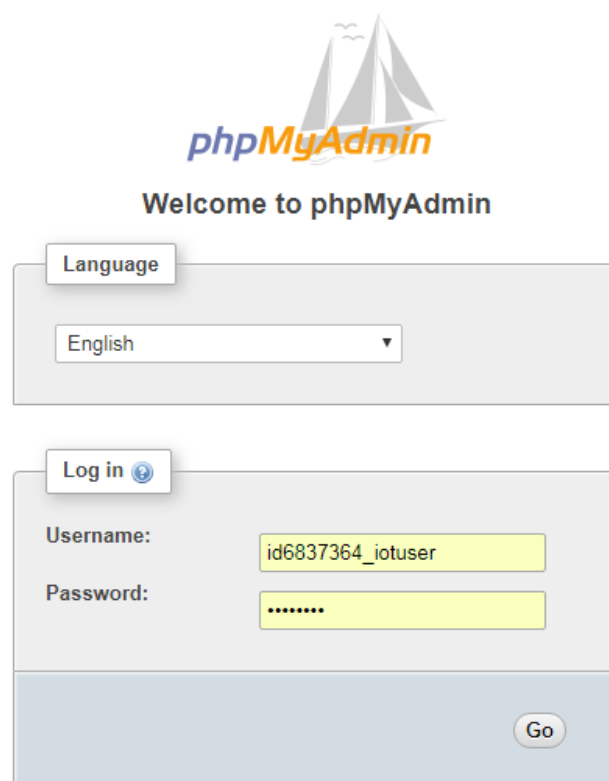
Cloud storage typically refers to a hosted object storage service, but the term has broadened to include other types of data storage that are now available as a service, like block storage.

Object storage services like Amazon S3, Oracle Cloud Storage and Microsoft Azure Storage, object storage software like Openstack Swift, object storage systems like EMC Atmos, EMC ECS and Hitachi Content Platform, and distributed storage research projects like OceanStore and VISION Cloud are all examples of storage that can be hosted and deployed with cloud storage characteristics.

For our project we use 000webhost.com as free web cloud server :-



PhpMyadmin : to create the databse and tables we are using phpMyadmin



← → ↻ https://databases.000webhost.com/sql.php?server=1&db=id6837364_iotdata&table=iottable&pos=0&token=b970655b71c9eeb3d0db002e6c145fa6

phpMyAdmin

Recent Favorites

- New
- id6837364_iotdata
 - New
 - iottable
 - Columns
 - New
 - status
 - time
- information_schema
- mysql

Server: localhost » Database: id6837364_iotdata » Table: iottable

Browse Structure SQL Search Insert Export Import Operations Triggers

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available. ⓘ

✓ Showing rows 0 - 17 (18 total, Query took 0.0127 seconds.)

```
SELECT * FROM `iottable`
```

☐ Show all | Number of rows: 25 ▾ | Filter rows:

+ Options

time status

5.1.2 Programming:

```
#include <ESP8266WiFi.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
#include "DHT.h"
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
#define DHTPIN D4 //connect DHT data pin to D4
#define DHTTYPE DHT11 // DHT 11
DHT dht(DHTPIN, DHTTYPE);

#define Relay1      D5
#define Relay2      D6
#define Relay3      D7
#define soil        A0

#define WLAN_SSID   "iotcold"           // Your SSID
#define WLAN_PASS   "1234567890"       // Your password

/***** Adafruit.io Setup *****/

#define AIO_SERVER   "io.adafruit.com" //Adafruit Server
#define AIO_SERVERPORT 1883
#define AIO_USERNAME "sk7030"          // Username
#define AIO_KEY      "aio_oUiA289ShSgiRHCZ0iLR7aOOd1nn" // Auth Key

//WIFI CLIENT
```

```
WiFiClient client;
```

```
Adafruit_MQTT_Client mqtt(&client, AIO_SERVER, AIO_SERVERPORT,  
AIO_USERNAME, AIO_KEY);
```

```
Adafruit_MQTT_Subscribe Light1 = Adafruit_MQTT_Subscribe(&mqtt,  
AIO_USERNAME"/feeds/AC"); // Feeds name should be same everywhere
```

```
Adafruit_MQTT_Subscribe Light2 = Adafruit_MQTT_Subscribe(&mqtt,  
AIO_USERNAME"/feeds/FAN"); // Feeds name should be same everywhere
```

```
Adafruit_MQTT_Subscribe Light3 = Adafruit_MQTT_Subscribe(&mqtt,  
AIO_USERNAME"/feeds/GAS"); // Feeds name should be same everywhere
```

```
Adafruit_MQTT_Publish Temperature = Adafruit_MQTT_Publish(&mqtt,  
AIO_USERNAME "/feeds/TEMP");
```

```
Adafruit_MQTT_Publish Humidity = Adafruit_MQTT_Publish(&mqtt,  
AIO_USERNAME "/feeds/HUM");
```

```
Adafruit_MQTT_Publish Gaslevel = Adafruit_MQTT_Publish(&mqtt,  
AIO_USERNAME "/feeds/GASV");
```

```
float temp; //to store the temperature value
```

```
float hum; // to store the humidity value
```

```
int value;
```

```
void MQTT_connect();
```

```
void setup() {
```

```
  Serial.begin(115200);
```

```
  lcd.init();
```

```
  lcd.backlight();
```

```
  pinMode(Relay1, OUTPUT);
```

```

pinMode(Relay2, OUTPUT);
pinMode(Relay3, OUTPUT);

// Connect to WiFi access point.
Serial.println(); Serial.println();
Serial.print("Connecting to ");
Serial.println(WLAN_SSID);

WiFi.begin(WLAN_SSID, WLAN_PASS);
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println();

Serial.println("WiFi connected");
Serial.println("IP address: ");
Serial.println(WiFi.localIP());

mqtt.subscribe(&Light1);
mqtt.subscribe(&Light2);
mqtt.subscribe(&Light3);

}

void loop() {

```



```

MQTT_connect();
  lcd.setCursor(0,0);
  lcd.print("T:");
  lcd.setCursor(2,0);
  lcd.print(temp);
  lcd.setCursor(8,0);
  lcd.print("H:");
  lcd.setCursor(10,0);
  lcd.print(hum);
  lcd.setCursor(0,1);
  lcd.print("Gas:");
  lcd.setCursor(5,1);
  lcd.print(value);

```

```

Adafruit_MQTT_Subscribe *subscription;
while ((subscription = mqtt.readSubscription(20000))) {
  if (subscription == &Light1) {
    Serial.print(F("Got: "));
    Serial.println((char *)Light1.lastread);
    int Light1_State = atoi((char *)Light1.lastread);
    digitalWrite(Relay1, Light1_State);
  }

```

```

    if (subscription == &Light2) {
      Serial.print(F("Got: "));
      Serial.println((char *)Light2.lastread);
      int Light2_State = atoi((char *)Light2.lastread);

```

```

    digitalWrite(Relay2, Light2_State);
}

    if (subscription == &Light3) {
    Serial.print(F("Got: "));
    Serial.println((char *)Light3.lastread);
    int Light3_State = atoi((char *)Light3.lastread);
    digitalWrite(Relay3, Light3_State);
    }
}

    value = analogRead(soil);
    value = map(value, 0, 1024, 0, 100);
    value = (value - 100) * -1;

    temp = dht.readTemperature();

    hum = dht.readHumidity();
    Serial.print("temperature = ");
    Serial.println(temp);
    Serial.print("humidity = ");
    Serial.println(hum);

    delay(10000);

    if (! Temperature.publish(temp)) {                //Publish to Adafruit
        Serial.println(F("Failed"));
    }

```

```

        if (! Humidity.publish(hum)) {                //Publish to Adafruit
            Serial.println(F("Failed"));
        }
        if (! Gaslevel.publish(value)) {              //Publish to Adafruit
            Serial.println(F("Failed"));
        }
        else {
            Serial.println(F("Sent!"));
        }
        lcd.clear();

    }

void MQTT_connect() {
    int8_t ret;

    if (mqtt.connected()) {
        return;
    }

    Serial.print("Connecting to MQTT... ");

    uint8_t retries = 3;

    while ((ret = mqtt.connect()) != 0) {
        Serial.println(mqtt.connectErrorString(ret));
        Serial.println("Retrying MQTT connection in 5 seconds...");
        mqtt.disconnect();
        delay(5000);
    }
}

```

```
    retries--;  
    if (retries == 0) {  
        while (1);  
    }  
}  
Serial.println("MQTT Connected!");  
  
}
```

CHAPTER 6

Implementation

CHAPTER No. 6

IMPLEMENTATION

6.1 Pictorial View of the Project

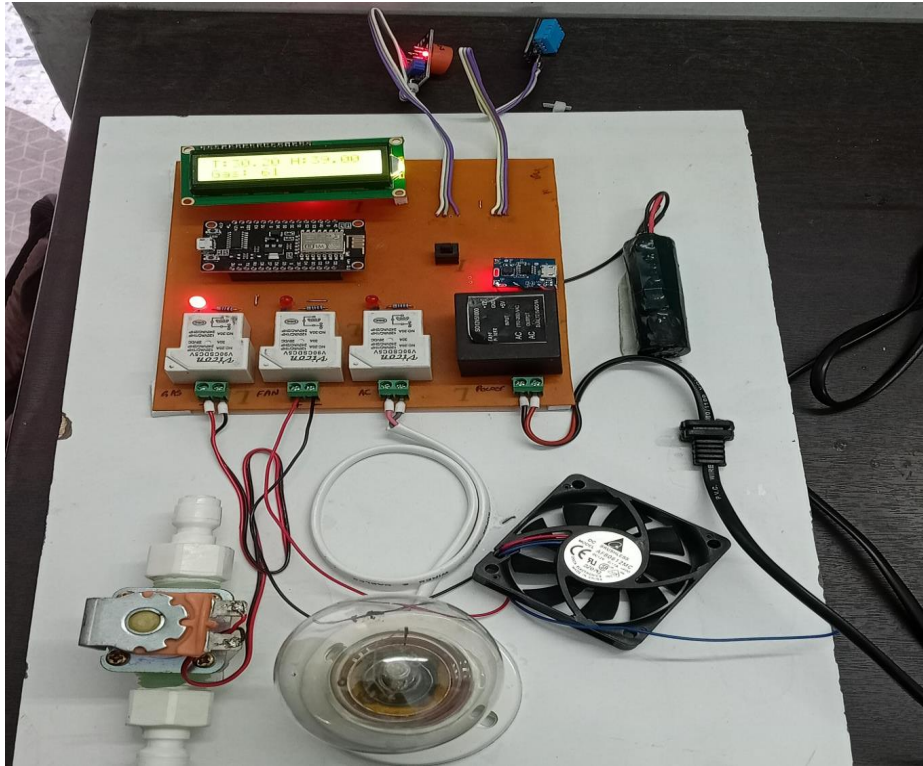


Figure No 5.1 Project Model Photo

The implementation of an IoT-based cold storage monitoring and controlling system involves several steps, including:

1. Planning:-

The first step is to conduct a feasibility study and determine the requirements of the system. This includes identifying the sensors and IoT devices needed, the communication network, and the software required to collect and analyze the data.

2. Hardware Installation:-

The next step is to install the hardware components of the system, including the sensors, actuators, and microcontrollers. The sensors should be placed strategically throughout the cold storage facility to ensure accurate reading.

3. Software Development:-

The software required to collect and analyze the data should be developed. This includes developing an interface that allows users to monitor and control the cold storage environment.

4. Testing:-

The system should be tested to ensure that it is functioning properly. This includes testing the sensors, communication network, and software. Any issues or faults identified during testing should be addressed before the system is deployed.

5. Deployment:-

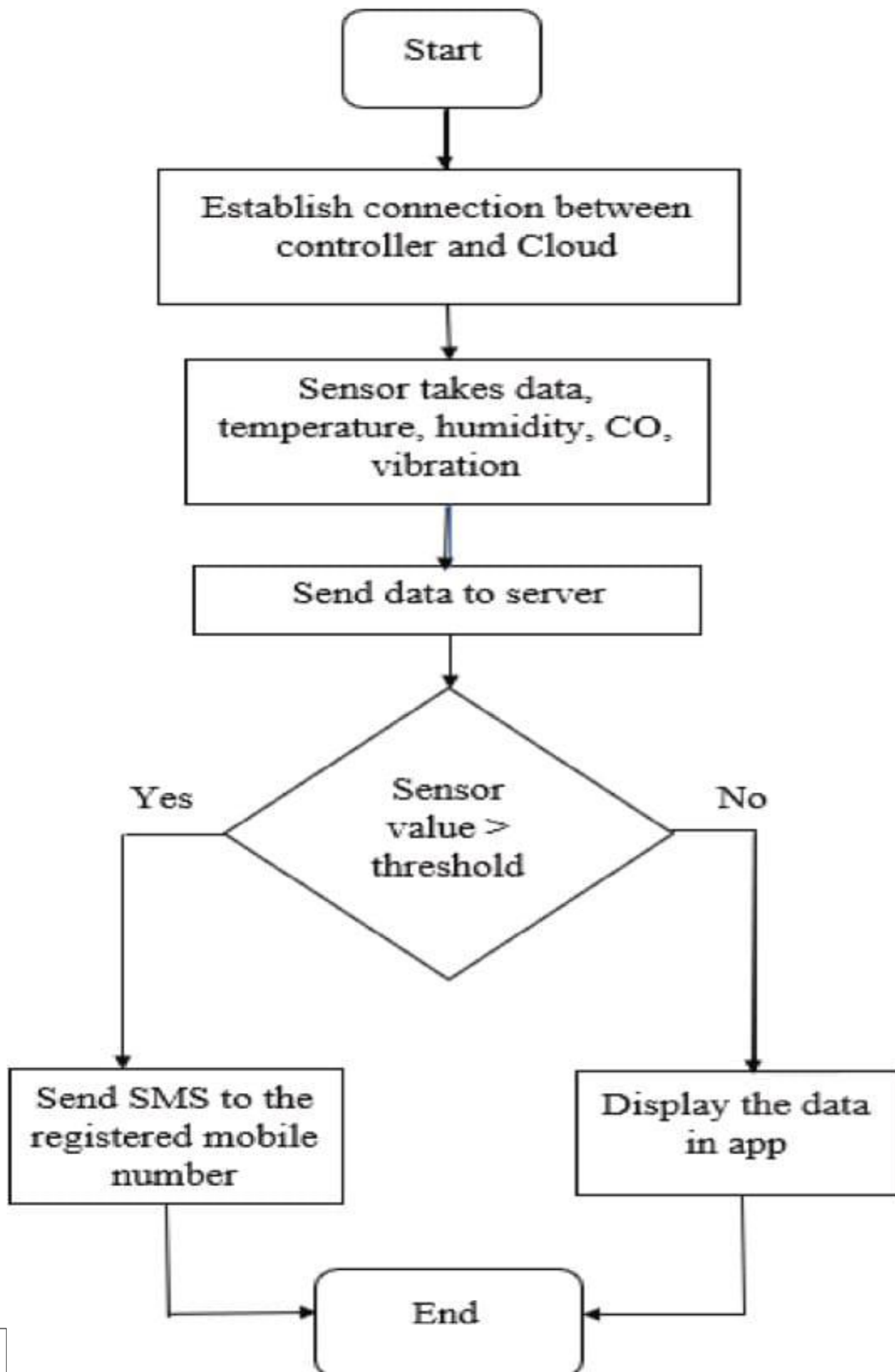
Once the system has been tested and is functioning properly, it can be deployed. The system should be integrated with the existing cold storage infrastructure, and employees should be trained on how to use the system.

6. Maintenance:-

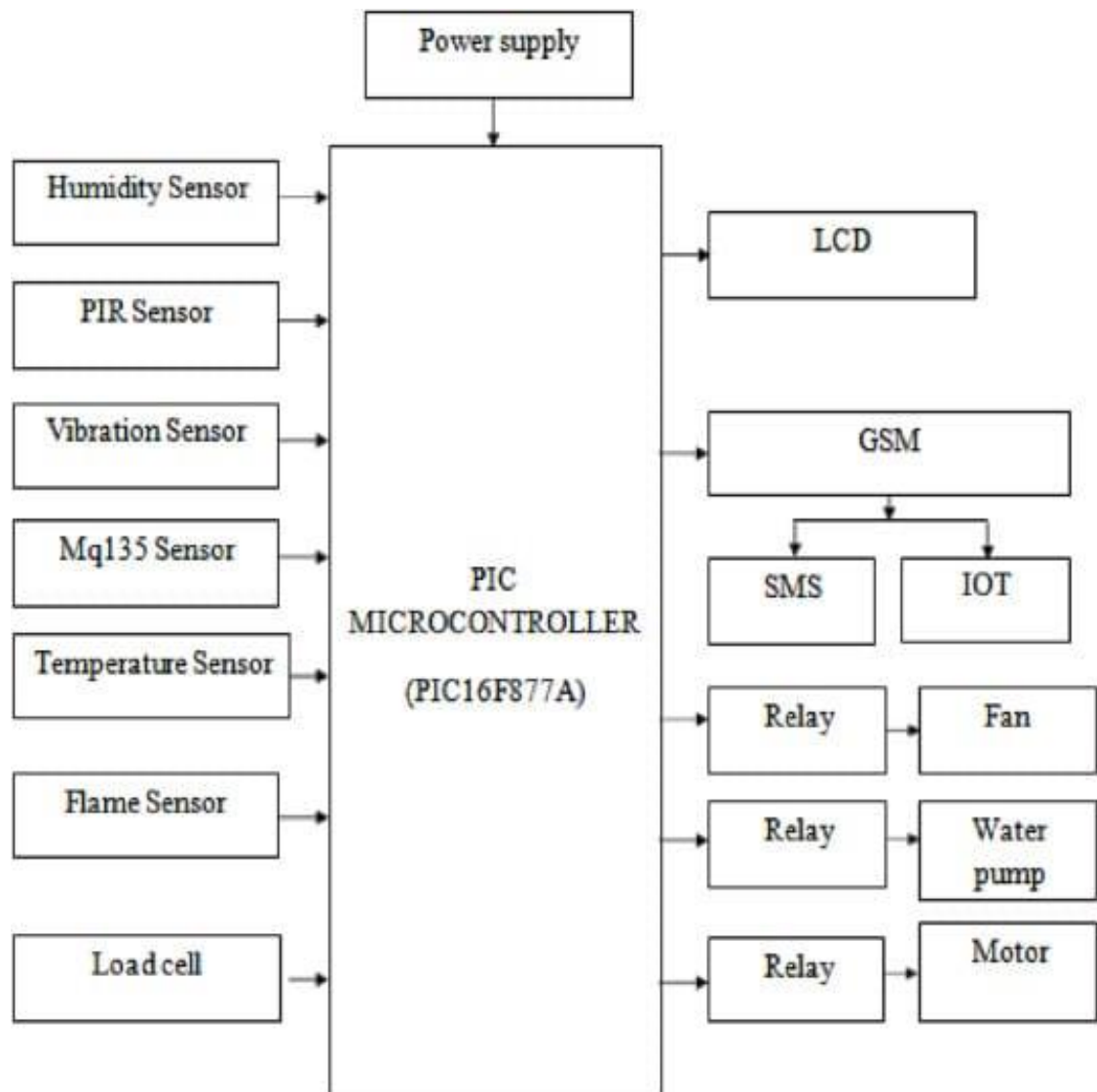
Regular maintenance of the system should be carried out to ensure that it is functioning optimally. This includes calibrating the sensors, updating the software, and replacing any faulty components.

Overall, the implementation of an IoT-based cold storage monitoring and controlling system requires careful planning and execution. With proper implementation, the system can help businesses improve efficiency, reduce waste, and enhance food safety.

6.2. Flowchart of Procedure to Obtain the Result:-



6.3. Architecture of IOT based food cold storage monitoring and controlling system:-



6.4. APPLICATIONS:-

An IoT-based cold storage monitoring and controlling system can have several applications in various industries, including:

6.4.1.Food and Beverage Industry:

In this industry, the IoT-based cold storage system can help in monitoring the temperature and humidity levels of the cold storage units, which is crucial for preserving the quality and freshness of food products. The system can also provide alerts and notifications in case of any fluctuations in temperature or humidity levels.

6.4.2.Pharmaceutical Industry:

The IoT-based cold storage system can help in monitoring the temperature and humidity levels of the cold storage units, which is crucial for preserving the quality and efficacy of drugs and vaccines. The system can also provide alerts and notifications in case of any fluctuations in temperature or humidity levels.

6.4.3.Agriculture Industry:

In this industry, the IoT-based cold storage system can help in monitoring the temperature and humidity levels of the cold storage units used for storing fruits and vegetables. The system can also provide alerts and notifications in case of any fluctuations in temperature or humidity levels, which can help in preventing spoilage and wastage of produce.

6.4.4. Chemical Industry:

The IoT-based cold storage system can help in monitoring the temperature and humidity levels of the cold storage units used for storing chemicals and other hazardous materials. The system can also provide alerts and notifications in case of any fluctuations in temperature or humidity levels, which can help in preventing accidents and ensuring safety.

Overall, an IoT-based cold storage monitoring and controlling system can help in ensuring the quality, safety, and efficiency of cold storage operations in various industries.

CHAPTER 6

Result and Conclusion

CHAPTER No. 6

RESULT AND CONCLUSION

6.1.1. Result :-

As a result, the effort required to supervise manual employees can be significantly decreased, thus enhancing and guaranteeing occupational health and safety. Traditional methods, except for temperature and relative humidity, do not provide real-time monitoring of various environmental data. Additionally, there is no system in place for updating the staff on the status of a commodity in cold storage in real-time. To prevent the loss of perishable goods in cold storage, we have demonstrated an intelligent system for real-time monitoring and alerting of cold storage that is based on an IoT-enabled strategy. This system automatically monitors and notifies users of dangerous environmental parameter thresholds like temperature, relative humidity, gas level, and light intensity. Additionally, presented data, including temperature, gas level, and humidity, will be displayed on the website we designed as well as on the LCD screen. The management of products kept in the warehouse is the purpose of this project. The business model is tailored to the convenience and eating preferences of the preceding generation's rural and urban residents. Frozen food's safety and nutritional worth should only be highlighted when quality raw materials are utilized, good production processes will be applied throughout the preservation process, and the items will be stored at preset temperatures.

6.1.2. Conclusion:-

We proposed and developed a prototype of an efficient, price-powerful, and accurate smart cold garage device with the usage of ESP32. Our clever bloodless garage machine is extra accurate and is capable of notifying the details about the contents of the garage. Temperature, pressure, gas value, and humidity inside the storage device may also be indicated within the distinctive app via the consumer's smartphone. For this reason, details of the stock inside the garage can be done without difficulty tracking, and refilling of the items may be finished via the consumer

CHAPTER 7

Future Scope

7.1. Limitation:-

There are several limitations of an IoT-based cold storage monitoring and controlling system, including:

1. Dependence on Internet Connectivity:

An IoT-based cold storage system relies heavily on internet connectivity. If there is no internet connection or if the connection is lost, the system will not function properly.

2. Security Risks:

IoT devices are vulnerable to cyber-attacks, and a cold storage system that relies on IoT devices is no exception. Hackers can exploit vulnerabilities in the system and gain access to sensitive data or take control of the system.

3. Cost:

Implementing an IoT-based cold storage system can be expensive, especially for small businesses. The cost of the hardware, software, and maintenance can add up quickly.

4. Maintenance:

IoT devices require regular maintenance to ensure that they are functioning properly. This can be time-consuming and costly.

5. Compatibility:

IoT devices from different manufacturers may not be compatible with each other, which can make it difficult to integrate them into a single system.

6. Data Overload:

IoT devices generate a lot of data, and it can be challenging to manage and analyze all of this data effectively.

7. Power Outages:

Power outages can disrupt the functioning of an IoT-based cold storage system, which can lead to spoilage of goods.

7.2 Future Scope:-

1. Implementing AI algorithms to analyze real-time data from sensors for predictive maintenance and optimization of cold storage operations.
2. Utilizing blockchain to create immutable records of temperature and humidity data, enhancing transparency and traceability throughout the cold storage supply chain.
3. Deploying edge computing devices to process data locally in the cold storage facility, reducing latency and improving responsiveness for critical temperature control actions in real-time.

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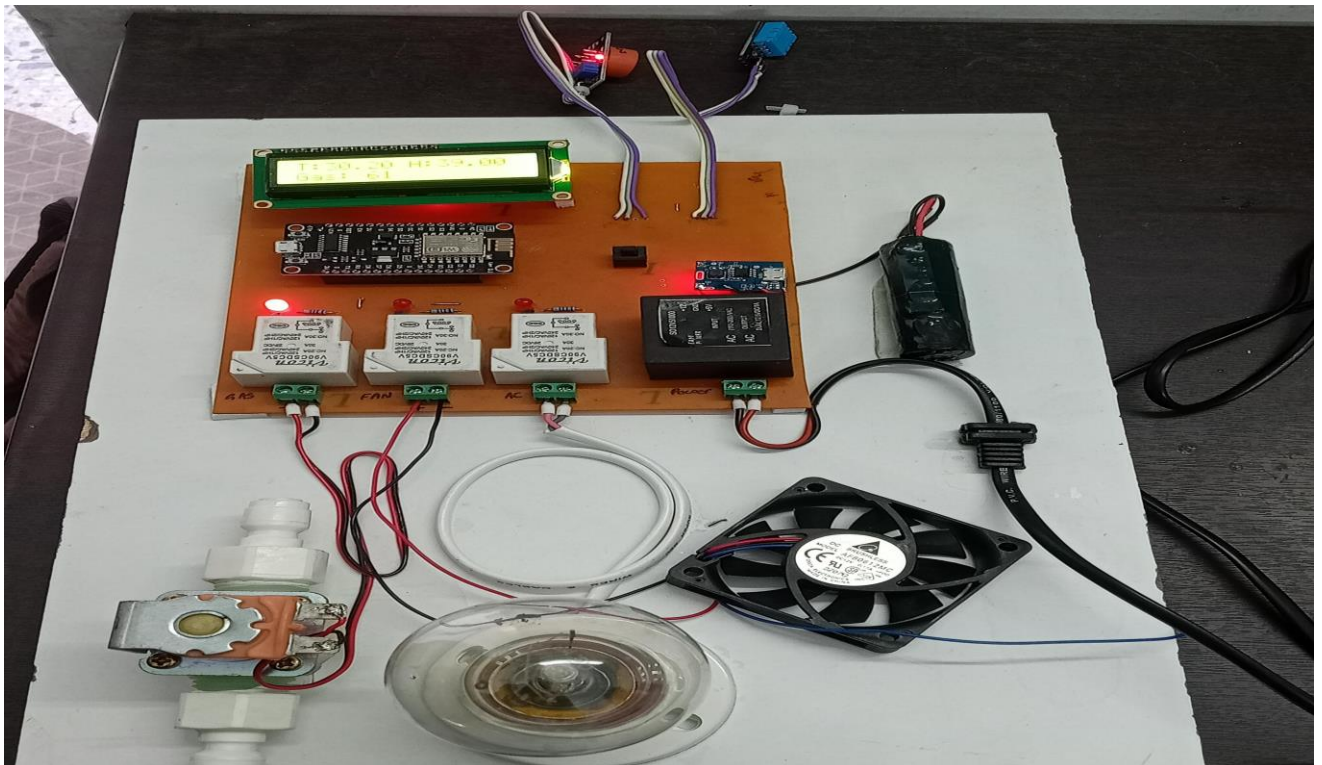
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ANNEXURE

PROJECT GROUP PHOTO

+ ADD GROUP PHOTO WITH PROJECT GIUDE



PUBLICATIONS

1. Ambika Nanwatkar, Sahil Jawade, Sagar Hiwaral, Vijay Gaikwad, Nikita Paul, Senha Parbat, Lalita Patle, and Prof. Nikhita Khobragade. "Project Title :- IOT based food cold storage monitoring and controlling system" International Journal of Advanced Research in Computer and Communication Engineering

Impact Factor 8.102, Peer-reviewed & Refereed journal Vol. 13, Issue 3, March 2024

DOI: 10.17148/IJARCCE.2024.13316

Project Completion:-

1. Participated in project competition in Guru Nanak institute of engineering and technology

PAPER



“IOT BASED FOOD COLD STORAGE MONITORING AND CONTROLLING SYSTEM”

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Abstract: Tricholoma matsutake (T. matsutake) is a special type of fungus known as “the king of bacteria”, and has a very high economic value. However, it is also very difficult to transport due to its corruptibility. Therefore, tracing and tracking the quality and safety of T. matsutake in the cold chain is very important and necessary. Based on changes in the cold chain, environmental parameters determine the safety of T. matsutake is a viable option. This paper developed and tested a real-time monitoring traceability system (RM-TM) using emerging Internet of Things (IoT) technologies for monitoring the cold chain logistics environmental parameters of T. matsutake. Finally, system testing and evaluation have shown that RM-TM can track and monitor temperature, humidity, oxygen and carbon dioxide fluctuations in the cold chain in real-time. In addition, the collected data can be used to increase the transparency of cold chain logistics and to more effectively control quality, safety, and traceability. In general, the system evaluation results show that it is reliable and meets the requirements of users.

In the Energy Management system, the main constraints are accurate data, energy monitoring and implementation of visual data for consumers. This Project is intended in designing a system at home or industry which monitors the temperature consumption of cold storage, which is designed to calculate the total energy consumption. A server will be created with appropriate channels to monitor the energy consumption from each of the devices respectively. These data will be uploaded to the server at the monitoring end. Considering all this data, an individual energy load profile for each of the devices is displayed on the web-page.

Keywords: Transmitter, Receiver, Antenna, Fading, Peak to average power ratio (PAPR), Bit error rate (BER), Symbol error rate (SER), Frame error rate (FER), Inter carrier interference (ICI), Inter symbol interference (ISI), Cyclic prefix (CP), Maximal ratio combining (MRC), Maximum sum rate and Minimum error rate etc.

I. INTRODUCTION

The “IOT BASED FOOD COLD STORAGE MONITORING AND CONTROLLING

SYSTEM” project revolutionizes the cold storage industry by introducing an automated solution to ensure the consistent preservation of frozen food. With the increasing demand for quality and safety in food storage, the system addresses crucial challenges by leveraging IoT technology. By controlling equipment like solenoid valves and air conditioners based on preset timers, the system minimizes human intervention, reducing the risk of temperature fluctuations that could lead to spoilage. Furthermore, real-time monitoring of environmental parameters such as temperature and humidity adds an extra layer of precision, enhancing food safety protocols and enabling efficient energy management. As a result, this project not only streamlines operations in cold storage facilities but also contributes to minimizing food wastage and optimizing resource utilization.

Incorporating cutting-edge IoT technologies, the “IOT BASED FOOD COLD STORAGE MONITORING AND CONTROLLING SYSTEM” project provides a comprehensive solution for the challenges faced by the cold storage industry. By automating temperature control processes and offering real-time monitoring capabilities, the system ensures the preservation of frozen food at optimal conditions. Additionally, the integration of cloud-based data storage facilitates remote access and management, enabling stakeholders to monitor operations from anywhere. This project not only enhances food safety and quality but also promotes sustainability by reducing energy consumption and minimizing food wastage, marking a significant advancement in the efficiency and effectiveness of cold storage operations.



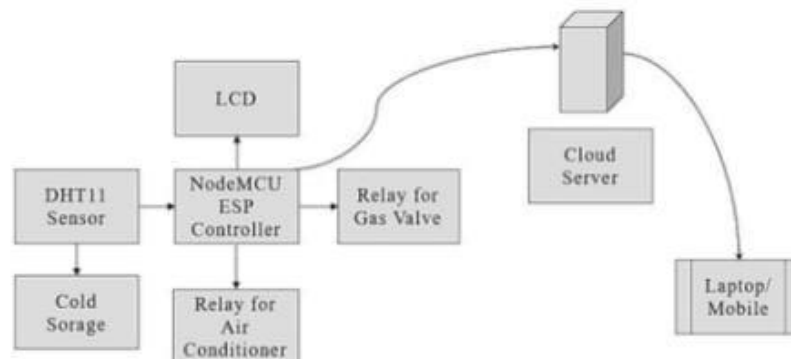
II. PROJECT OBJECTIVE

1. To control solenoid valve for ammonia gas
2. To control air conditioner unit
3. As per timer, both systems should work
4. Both systems manage online – It has timer values to set it ON and as per timer over the system will turn OFF
5. Temperature and humidity parameters should monitor online – If power cut occur time should be noted till the power
6. Data must be seen on the cloud server

III. LITERATURE REVIEW

The literature survey encompasses various studies highlighting innovative approaches in temperature monitoring and food storage management. Budijono and Felita's study emphasizes the significance of temperature control in food quality maintenance, employing ESP32 to monitor freezer performance continuously. The integration of Wi-Fi connectivity enables real-time data transmission to a cloud database, facilitating remote monitoring and preventive maintenance measures. Additionally, Bamanikar et al.'s exploration of smart refrigerators underscores the transformation of conventional refrigeration units into IoT-enabled systems, enhancing user accessibility and food preservation efficacy. Similarly, Murali et al.'s IoT-based smart refrigerator introduces automated inventory management through weight detection, streamlining grocery ordering processes and optimizing resource utilization. Witjaksono et al.'s investigation focuses on utilizing IoT for food quality and safety monitoring, emphasizing the potential of smartphone applications in disseminating real-time food condition data to consumers and supervisors. These studies collectively underscore the evolving landscape of food storage technologies, emphasizing the integration of IoT for enhanced efficiency, convenience, and food safety.

IV. BASIC BLOCK DIAGRAM



1. For monitoring the temperature DHT11 temperature sensor is used.
2. Temperature sensor is connected to the ESP Controller.
3. Temperature sensor records the data which is stored on cloud server to monitor with respect to date and time as well as in the form of graph. So, the temperature of cold storage is continuously monitored through IoT.
4. Cloud server provides wireless communication which is also connected to ESP Controller.









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INDUSTRY TIE-UP LETTER



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Date:-

Name Of Student:- Mahesh Raju Verma

At-Salori, Post-Chinora
Tah-warora, Dist.Chandrapur
Pin-442907

Subject:- Letter Of Internship Acceptance

Dear Mahesh Verma

We Pleased to offer you an internship program with Sai Wardha Energy, Warora. Period of 6 months on our upcoming project "Condition Monitoring Motor" using AT.

You should note that any information data collect form during the course of your internship should be kept confidential at all time. Appreciate your intrest for our project and us.

Your Sincerely

An ISO 9001:2008, ISO 14001:2004 & OHSAS 18001:2007 Certified Company

Member  Williamson Magor Group

Corporate & Registered Office 4 Mangoe Lane Kolkata-700 001

